	NUMALIGARH REFINERY LIMITED	Total pages: 99

NRL EXPANSION PROJECT

PROCESS DESIGN BASIS OVERALL

Written By Karthikeyan Rajendran 2021.11.12 17:55:29 +05'30'

Rev.	Date	Reason for Issue	Prepared	Checked	Approved	Prepared	Review	Review
D4	12/11/2021	APPROVED FOR DESIGN	RK	SCB	AR			
D3	22/03/2021	APPROVED FOR DESIGN	RK	SCB	AR			
D2	12/10/2020	APPROVED FOR DESIGN	RK	SCB	AR			
D1	07/09/2020	APPROVED FOR DESIGN	RK	SCB	AR			
A1	19/06/2020	ISSUED FOR REVIEW	RK	SCB	AR			
			Discipline Engineer	Discipline Lead	Contractor Representative	Discipline Engineer	Project Engineer	Department Head
			TPIL			**NRL**		

TECHNIP INDIA LIMITED

	Category	Code	Description
	Facility Area Code	1ZZZA	Common Document
NRL's PO NO: 4300062833-AMA/16.06.2020	Document Type	BOD	Basis of Design
TP DOC REF: 082176C-ZZZ-CN-0007-0001_D4	System Number	00	General
	Life Cycle	01	Disk Ref.:

Originator/ Contractor	Asset Code	Discipline	Document Type	Sequence Number	Revision
TP	**1ZZZA**	**PR**	**BOD**	**0001**	**D4**

TABLE OF CONTENTS

ATTACHMENTS

Attachment 1 – Basic Engineering Design Data (14 Pages)

REFERENCES

Engineering Tag Number Specification, Document number NR-0ZZZZ-GE-SPE-0002

1. INTRODUCTION

NUMALIGARH REFINERY LIMITED (NRL) is a Government of India Enterprise located at Numaligarh, Dist. Golaghat, Assam. At present, Numaligarh refinery is processing 3 MMTPA of Assam Mix Crude Oil. Existing configuration includes CDU/VDU, DCU, HCU, HGU, DHT, MSU, Coke calcination unit, SRU and Wax unit apart from the Marketing terminal and LPG Bottling plant.

Numaligarh Refinery Limited is expanding its refining capacity from current 3 MMTPA to 9 MMTPA by installing a new 6 MMTPA capacity CDU/VDU and associated downstream process units, utility & offsites, as part of Numaligarh Refinery Expansion Project (NREP).

Numaligarh Refinery Limited (NRL) has selected M/s. Technip India Limited (TPIL) to provide M-PMC / EPCM services and supply of Basic Process Package for CDU/VDU with (ATU), ARU & SWS units. M-PMC (Managing PMC) shall be the overall coordinator for the entire project and also act as EPCM for LPGTU and Utility & Offsite facilities. M-PMC shall be responsible for entire NREP activities in respect to integration and schedule completion including interface management with EPCM-1, EPCM-2 and BOO contractor.

2. DEFINITIONS AND ACRONYMS

Wherever the following terms used in this document shall have the meaning as defined below:

NRL or OWNER or CLIENT	shall mean Numaligarh Refinery Ltd.
M-PMC or PMC	shall mean TECHNIP India Ltd (TPIL).
LICENSOR	shall mean agency responsible for the supply of basic know-how and process package for the licensed units.
PROCESS DESIGNER	shall mean the agency responsible for supply of basic know-how and process package for the open art units / facilities.
EPC	shall mean the work for the Engineering, Procurement and Construction of the Project.
CONTRACT	shall mean the Contract including all Annexes thereto and all documents therein attached and amendments which the parties may hereafter agree in writing to be made to this Contract.
CONTRACTOR	shall mean any third party whose services are obtained for supply execution and / or erection of units/ facilities covered under contract scope.
VENDOR / SUPPLIER	shall mean any third party supplying any of the equipment / materials for setting up Plant covered under Scope of Work.
PROJECT	shall mean the Capacity Expansion of Numaligarh

	Refinery Project.
SITE	shall mean Numaligarh Refinery (Numaligarh, Dist. Golaghat, Assam, India).
PLANT	shall mean the units and facilities comprised in the Project, and if divided into different packages for the award of LSTK Contracts / EPCM Contracts / Work Contracts / Supply Contracts.
UNIT	shall mean a particular process unit etc. which forms a distinct operating system and is a part of the Plant.
BATTERY LIMIT	shall mean the demarcated area within which all the units and facilities under scope as detailed in the contract are being installed for Plant.

Wherever the following acronyms used in this document shall have the meaning as defined below:

AC	Alternating Current
ANSI	American National Standards Institute
APH	Air Preheater
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ATU	Amine Treatment Unit
BDEP	Basic Design and Engineering Package
BEDD	Basic Engineering Design Data
BFW	Boiler Feed Water
BL	Battery Limit
CBD	Closed Blow Down
CCRU	Continuous Catalytic Regeneration Reformer Unit
CDU	Crude Distillation Unit
CPCB	Central Pollution Control Board
CPU	Condensate Polishing Unit
CR-LPGTU	Cracked LPG Treatment Unit
CS / SS	Carbon Steel / Stainless Steel
CSO	Car Seal Open
CSC	Car Seal Closed
DCS	Distributed Control System
DFR	Detailed Feasibility Report
DHT	Diesel Hydrotreater Unit
DM	Demineralized
EPC	Engineering, Procurement, and Construction
EPCM	Engineering, Procurement, Construction, Management
ESD	Emergency Shutdown System
ETP	Effluent Treatment Plant
FD	Forced Draft
FEED	Front End Engineering Design
FG	Fuel Gas
FLO	Flushing Oil

GDS	Gasoline Desulfurization Unit
HAS	High Alloy Steel
HFLO	Heavy Flushing Oil
HGU	Hydrogen Generation Unit
HMB	Heat Material Balance
HP	High Pressure
IBR	Indian Boiler Regulation
ID	Induced Draft
ISA	International Society of Automation
ISBL	Inside Battery Limit
ISOM	Isomerization Unit
KL	Kiloliters
KOD	Knock Out Drum
KCS	Killed Carbon Steel
LAS	Low Alloy Steel
LP	Low Pressure
LPG	Liquified Petroleum Gas
LSTK	Lump Sum Turn Key
MDMT	Minimum Design Metal Temperature
MFB	Minimum Flow Bypass
MGD	Millions of Gallons per Day
MINAS	The Minimal National Standards
MMTPA	Million Metric Tons Per Annum
MOC	Material of Construction
MOEF & CC	Ministry of Environment & Forests and Climate Change
MOV	Motor Operated Valve
MP	Medium Pressure
M-PMC	Managing Project Management Consultant
MSP	Motor Spirit Plant
MT	Metric Tonnes
NACE	National Association of Corrosion Engineers
NHT	Naphtha Hydrotreating Unit
NREP	Numaligarh Refinery Expansion Project
NRL	Numaligarh Refinery Ltd
NSU	Naphtha Splitter Unit
OISD	Oil Industry Safety Directorate
OSBL	Outside Battery Limit
OWS	Oily Water Sewer
PESO	Petroleum & Explosives Safety Organization
P&ID	Piping & Instrumentation Diagram
PDS	Process Datasheet
PFCC	Petro Fluid Catalytic Cracking Unit
PLC	Programmable Logic Controller
PMC	Project Management Consultant
PMO	Project Management Office
PMS	Piping Material Specification
PP	Polypropylene Unit
PRU	Propylene Recovery Unit
PSA	Pressure Swing Adsorption

PSV	Pressure Relief Valve
PWHT	Post Weld Heat Treatment
RP	Recommended Practice
RPTU	Resid Processing & Treating Unit
SPM	Suspended Particulate Matter
SRB	Sulphur Recovery Block
SR-LPGTU	Straight Run LPG Treating Unit
TBD	To be defined
TEMA	Tubular Exchanger Manufacturers Association
TMTPA	Thousand Metric Tonnes per Annum
TPD	Tons per Day
TPH	Tons per Hour
TPIL	Technip India Ltd
TSO	Tight Shut Off
VDU	Vacuum Distillation Unit
VFD	Variable Frequency Drive

3. PURPOSE

The purpose of this document is to provide the General Requirements, Basic Process Design Requirements and Criteria, which have to be followed to perform the Process Engineering Design of Process (Open-art / Licensed) units, Utility units, Interconnecting and off site facilities for Numaligarh Refinery Expansion Project.

The objective of this document is to ensure the consistency in the design and the deliverables being part of the Numaligarh Refinery Expansion Project.

For licensed units, the rules of the process LICENSOR shall be used if they are more stringent than the rules provided in this document. In case some of the guidelines of the Process design basis are conflicting with other document, the order of priority is:

- National & Local Rules, Regulations, Codes & Standards and Statutory Requirements.
- Licensor Design Basis / Standards & Specification (Note 1)
- Project Standards & Specifications
- International Codes and Standards

Note 1:

In case of conflict between Licensor’s specification / Client (NRL) or M-PMC’s specification and Code requirements, the most stringent requirements of these shall govern with approval of Client (NRL) / M-PMC and the decision of Client (NRL) / M-PMC in resolution of the conflict shall be final.

The process design basis overall is meant mainly for grassroot / new units.

4. APPLICABLE CODES & STANDARDS & REGULATIONS

The Project shall comply with the requirements of Indian authorities as applicable including but not limited to the following

a. Factories Act
b. Indian Petroleum Rules
c. Tariff Advisory Committee Guidelines
d. Liquid effluent discharge, as per Minimal National Standards for liquid effluents and air Emissions conforming to pollution control board standards
e. Civil Aviation Rules
f. Indian Boiler Regulation Act
g. India Electricity Rules
h. Requirement of Chief Controller of Explosives (PESO)
i. Requirements of other authorities concerned with the Project or out of any license, permission, sanction, approval or no objection relative thereto
j. CEA (Central Electrical Authority)
k. Legal Metrology
l. CPWD standard shall be considered for civil material related testing
m. International Standards: ANSI, ASME, ASTM, API, SA, NACE, NFPA, ISO, DIN, EN etc.
n. All statutory provisions of India / State Govt.
o. Indian Oil Industry Safety Directorate (OISD) codes and standards
p. Compliance to recommendations of Statutory committee report – M B Lal Committee Report

The Indian Oil Industry Safety Directorate (OISD) codes and standards, in particular related to equipment spacing, are mandatory and take precedence in the event of conflict with other project standards.

Bureau of Indian Standards (IS) are the governing Standards for both Civil and Electrical disciplines.

5. GENERAL INFORMATION

Language : English

5.1 Project Title

Project Name	Numaligarh Refinery Expansion Project (NREP)
Owner	Numaligarh Refinery Limited
Location	Numaligarh Dist. Golaghat, Assam

5.2 Plant Location

Refer Engineering Design Basis for complete details of plant location. This section presents brief details

Site Location:

State / Country	ASSAM / INDIA
Nearest Railway Station	GOLAGHAT
Nearest Town / City	NUMALIGARH
Nearest Airport	JORHAT
Nearest National Highway	NH-39
Nearest Port	KOLKATA

Source of Water:

Fresh water : Dhansiri River

6. PROCESS UNITS UTILITY & OFFSITE FACILITIES

Process Unit, Utility and Offsite Facilities covered under this project,

Process / Utility Unit Description	Unit Number	Capacity	Remarks
New Units			
Crude Distillation Unit (CDU), Vacuum Distillation Unit (VDU) with Amine Treating Unit (ATU)	CDU/VDU - 1P21 ATU – 1P34	6.0 MMTPA	
Naphtha Hydrotreater (NHT)	1P22	1.16 MMTPA	(Part of MSP)
Continuous Catalytic Reforming Unit	1P23	0.789 MMTPA	(Part of MSP)
ISOM	1P24	0.465 MMTPA	(Part of MSP)
Resid Processing & Treating Unit (RPTU)	RPTU - 1P30 VGO HDT – 1P31	2.0 MMTPA 2.27 MMTPA	
Diesel Hydrotreater (DHT)	1P28	3.55 MMTPA	
Petro Fluidized Catalytic Cracking (PFCC) / LPGTU-2	1P25 1P26	1.95 MMTPA / 793.55 KTPA	
Gasoline Desulfurization (GDS)	1P27	0.66 MMTPA	
H2 Generation Unit (HGU)	1P29	95 KTPA	
Sulphur Recovery Block (SWS/ARU/SRU/TGTU)	SWS-1P32, ARU-1P33, SRU 1 -1P35, SRU 2 - 1P36, TGTU-1P37	2 x 240 TPD	
LPGTU-1 (CDU-VDU / RPTU)	1P38	138.97 KTPA	

Offsite & Utilities (*)			
Raw Water Storage and Distribution	1U51	60000 m3 & 3400 m3/hr	
Drinking water system	1U52	130 m3/hr	
Cooling Water System	1U53	CT-1: 7 cells (6W+1S) CT-2: 7 cells (6W+1S) (4000 m3/hr per cell)	
DM/Process Water System	1U59	4 x 250 m3 /hr (3W + 1S Trains)	
Condensate Polishing Unit DM Water Tank	1U59	2 x 120 m3/hr (1W +1S Train) 2 x 7287 m3 **	
Steam Generation System	1U60	3 x 125 TPH (1W+1S) boilers, 1W HRSG	
Instrument Air/Plant Air System	1U54	4 x 90000 Nm3/h (3W+1S)	
Nitrogen generation plant	1U55	5000 Nm3/h	
Fuel gas system	1U58	TBD *	
Flare System (HC & Acid Gas)	1U61	TBD *	
Raw Water Intake Facility	1U66	3400 m3/hr	
Raw Water Desilting System	1U67	3700 m3/hr	
Crude Oil Storage Tank	1S71	3 x 35800 m3**	
RPTU Feed Tank	1S72	2 x 29783 m3**	(New)
HCGO Tank	1S72	1 x 3319 m3**	
DCU Feed Tank	1S72	2 x 6583 m3**	(New)
NHT feed Tank	1S72	2 x 9800 m3**	
SR VGO Tank	1S72	3 x 20700 m3**	
GDS Feed Tank	1S72	1 x 9800 m3**	
PFCC Feed Tank	1S72	2 x 17595 m3**	
DHT Feed Tank	1S72	2 x 19650 m3**	(New)
Propylene Mounded Bullets	TBD	4 x 3350 m3	(Future)
Isomerate / Reformate Tanks	1S72	4 x 5000 m3** 1 x 11500 m3** 1 x 15100 m3**	(Existing)
MS Tanks	1S73	3 x 29210 m3** 4 x 7900 m3**	(Existing) (Existing)

HSD Tanks	1S73	3 x 23100 m3** 1 x 20700 m3** 3 x 27700 m3**	(Existing)
Light Slop	1S72	2 x 5161 m3	(New)
Heavy Slop	1S72	2 x 5000 m3	(New)
Chemical storage / handling/ pump house	1S74	TBD	
Product blending facility	1S75	2 nos. for HSD / Gasoline	
LPG Storage and Disposal facility	1S76	6 x 4045 m3	(2 nos. future)
Sulfur storage & disposal facility	1S78	TBD	
New ETP	1S79	400 m3/h	

*- Capacity to be confirmed later.

**- Capacity indicated is Stored Capacity.

7. MECHANICAL DESIGN CONDITIONS

7.1 DESIGN TEMPERATURE

7.1.1 Equipment / Piping Operating at Temperature above 0 °C

As a general rule the design temperature shall be defined as follows:

$T_D = T_{MCO} + 15$ °C as minimum requirement

Design temperature shall not be lower than 65 °C
Shall be round up to the nearest 5°C above unless flange rating affected

Where:

T_D = Design temperature (°C)

T_{MCO} = Maximum continuous operating temperature (°C) is a maximum continuous operating temperature reached during an operating cycle (e.g. end of run condition). Its selection must take into account some specific operating conditions: turndown conditions, different feed and operating cases, etc.

For pressurized LPG storage facilities, the design temperature is equal to the corresponding boiling temperature at design pressure and assessed LPG composition.

Exceptional operating temperatures which correspond to alternate or transient operations, such as regeneration, dry out, or bypass of heat exchanger of a feed/effluent heat exchangers train have to be considered, the duration of the corresponding operations exceeding a total of 100 hours per year. If there is no change in operating pressure, the process design temperature will be the maximum of the two values:

- Either the maximum temperature for alternate transient operation
- Or the maximum continuous operating temperature + 15 °C minimum.

If there is a significant change in operating pressure for these exceptional operating conditions (for example reaction loops with in-situ catalyst regeneration, decoking conditions), another set of design temperature and pressure has to be specified corresponding to these operating conditions. Refer to section 7.1.2 special case.

The accidental temperatures, which can occur in emergency situations such as loss of utilities, valve failure or during start-up, shut-down or any abnormal operation corresponding to a short duration and are not taken into account as long as the temperature increase does not exceed codes limits (investigation has to be followed with specialists). However, some exceptions can be made to this general rule on a case to case basis as described in section 7.1.2 special case.

Equipment containing parts, which can be damaged by abnormal high temperature, has to be designed for this temperature. It concerns column internals, desalter internal electrical isolating material, heat exchanger or air coolers tubes with polymer coating. For this type of equipment, steam out delivery conditions have to be reconsidered in order to remain below the maximum acceptable temperature. (if steam-out conditions are governing then it will be reviewed on case by case basis).

Accidental temperatures, which can occur in fire case situation, will not be considered to define the Design Temperature as adequate mitigation measures (e.g. depressurizing system, firefighting) have to be considered.

For piping, accidental temperatures are not considered for piping class selection, but shall be indicated as flexibility temperature in the line list for pipe flexibility study.

7.1.2 Special Cases

Condensing service by air coolers (column overhead, air condenser in reaction loop etc.). Design temperature downstream the air cooler is equal to air condenser normal inlet temperature decreased by expected cooling due to natural draught through the air cooler.

For air cooler on final product cooling, a similar approach can be taken for the design temperature of downstream equipment. But the design temperature of off-site lines will not be increased beyond the application of the general rule.

If for process reasons, a specified maximum operating temperature must not be exceeded, this maximum possible operating temperature is used as design temperature (e.g. catalytic reforming reactors). However, the design conditions of such reaction loops are defined by the Licensor and have to be followed.

When maximum continuous temperature (T_{MCO}) is unknown with sufficient accuracy or can be sensibly affected by feed quality, the design temperature can be exceptionally considered with a margin of 30 °C.

Equipment whose thickness is designed for creep (e.g. steam cracker furnace tubes), the combination of the maximum pressure and the maximum temperature is in general not considered simultaneously.

For heat exchanger trains with by-pass of individual exchanger, the design temperature (hot side) of the downstream exchanger will be the normal operating temperature of the by-passed exchanger assuming that shells are by-passed piece by piece (simultaneous by-pass of several exchangers in series is not considered) unless hot fluid is bypassed automatically, same philosophy will be applied to a trim cooler downstream air fin.

7.1.3 Steam Out

The steam out conditions (using LP steam) are as follows:

- External Pressure : 0.5 kg/cm^2(a) Half vacuum at 170 °C
- Internal Pressure : 0.5 kg/cm^{2}g at 170 °C

Steam out conditions (using MP steam) are as follows:

- External Pressure : 0.5 kg/cm2(a) Half vacuum at 225 °C
- Internal Pressure : 0.5 kg/cm^{2}g at 225 °C

Typically LP steam shall be used for steam out operations, option of utilizing MP Steam for steam out operation to be decided case to case based on specific requirement.

7.1.4 Emergency Depressurizing

The exceptional cold temperature generated by depressurizing of equipment or complete system will be indicated with the related residual pressure in order to select the material accordingly.

Only depressurizing actions, clearly identified in the operating manual for process or safety reasons are considered. Accidental depressurizing, due to single control valve / safety valve failure or operator error for example are not taken into account. Indeed, these potential cases can be faced by:

- Closing the block valves associated to the control valve
- Clear instruction / procedure within the operating manual to avoid such operator error

The equipment has to be reheated and slowly re-pressurized according to a written procedure. When immediate re-pressurization is required, a second set of design pressure/design temperature shall be specified.

7.1.5 Vacuum Conditions

A specific design temperature will be associated to specified vacuum design pressure.

7.1.6 Equipment operating at a temperature below 0 °C

As a general rule, the design temperature shall be defined as follows:

$T_D = T_{MCO} - 10\ °C$ (3)

Where:

T_D = Design Temperature (°C)
T_{MCO} = Minimum continuous operating temperature (°C) taking into account minimum operating pressure except depressurization.

Notes:

(1) If a piece of equipment normally operated below 0 °C can be submitted to temperature above ambient during particular operating phases (dry out, start-up, regeneration, …) or accidental causes (except fire), a hot design temperature will be indicated with associated design pressure. Such a data can have an impact on insulation type selection.

(2) For depressurization, refer to section 6.1.4

(3) If such temperature cannot be reached due to intrinsic process limits, the lowest possible temperature will be selected.
(4) Equipment in stand-by at ambient low temperature that can be pressurized for process reasons shall be specified with a second set of design conditions.

7.1.7 Minimum Design Metal Temperature (MDMT)

Refer to site data for MDMT (Low ambient temperature) as per Attachment-1 Basic Engineering Design Data (BEDD).

7.1.8 Discontinuous Processes

For discontinuous / cyclic processes (for example molecular sieves, regeneration, decoking), different sets of design temperature / pressure shall be specified to ensure that all operating cases are covered. The magnitude and frequency of the temperature / pressure swing shall also be provided. Mixing between sets of extreme conditions of pressure and temperature shall not be considered.

For cyclic operation, cycle duration has to be specified.

7.2 DESIGN PRESSURE

7.2.1 General Rule

The operating pressure (OP) is the pressure applied to the inside of a pressure vessel during its normal, steady operation. The maximum operating pressure (MOP) is a maximum continuous operating pressure reached during an operating cycle (e.g. End-of-Run conditions).

Based on MOP, Design pressure (DP) for columns, vessels, heat exchangers, reactors, and lines shall be established as follows:

Maximum Operating Pressure	Design Pressure
Atmospheric Pressure (Pressure vessels)	0.5 kg/cm^2(g)
Vacuum	Full Vacuum (FV) and 3.5 kg/cm^2(g) minimum (1)
Between 0 and 10 kg/cm^2(g)	Max. operating pressure + 1 kg/cm^2 as minimum requirement (3.5 kg/cm^2(g) min.) (2)
Between 10 kg/cm^2 (g) and 35 kg/cm^2(g)	Max. operating pressure + 10 % or 2 kg/cm^2(g) minimum whichever is higher
Between 35 kg/cm^2 (g) and 70 kg/cm^2 (g)	Max. operating pressure + 10 %
Above 70 kg/cm^2 (g)	Max operating pressure + 5% min.

For LPG (or similar service) storage systems under pressure, the MOP is equal to the vapor pressure at the maximum operating temperature of the assessed LPG composition.

Notes:

(1) Full vacuum design conditions will be applied to equipment that fulfil one of the following:
- Normally operates under vacuum.
- Is subject to vacuum during transient operation.
- When the equipment utilizing steam during normal operation, e.g. steam stripping.
- Can undergo vacuum through the loss of heat input (to be studied case by case). Vacuum prevention systems are also acceptable.

Design pressure for equipment shall consider vacuum condition which can occur due to equipment elevation: an example is given by exchangers in cooling water service normally (provided with vacuum breaker).

(2) For equipment in equilibrium with flare, the design pressure is the flare design pressure. For equipment provided with safety valve discharging to a flare system, the design pressure cannot be lower than the flare network design pressure.

For equipment provided with safety valve discharging to atmosphere the design pressure is 2.5 Kg/cm^2(g).

(3) Short time conditions for line design, as per ANSI B31.3 and 31.4, can be applied, when:
- for pressure surges due to water hammer.
- for decoking operations.

(4) Alternate operations, such as regeneration, dry-out, start-up, shut-down etc. shall be considered in determining the design pressure.

7.2.2 Mechanical Design Pressure for Storage Tanks

Design pressure of storage tanks shall be in accordance with API-620, API-650 and API-2000 (latest edition), as appropriate. Blanketed storage tanks shall have a blanketing design pressure of 150 mmWC, unless a higher pressure is deemed necessary from process considerations.

The design pressure of blanketed storage tanks shall be determined to allow adequate overpressure for out-breathing and emergency relief devices. The blanketed tanks shall also be designed for 50 mmWC vacuum.

7.2.3 Design Pressure Profile for Columns & Systems

The design pressure at the bottom of a fractionation column will be determined as follows:

$$P_{DB} = P_{DT} + \Delta P_1$$

Where:

P_{DB} = Design pressure at the bottom (vapor phase)
P_{DT} = Design pressure at the top calculated as per paragraph
$\Delta P1$ = Column pressure drop and liquid head (HHL) at column bottom

The liquid flowing density and the maximum liquid height will be indicated on the Process Data Sheet. Liquid head has to be considered in defining the design pressure of natural circulation reboilers.

7.2.4 Design Pressure at Discharge of Pump

In principle design pressure for lines and equipment located on the pump discharge shall be equal to pump process design pressure down to the last block valve located before a section protected by a safety valve.

The design pressure at the discharge of the pump will be set as follows, before pump selection:

$$DP_{discharge} = DP_{suction} + \frac{K \times \Delta H_{design} \times density_{max}}{A}$$

$DP_{discharge}$	=	Design pressure at pump discharge
$DP_{suction}$	=	Design pressure at pump suction
ΔH_{design}	=	Differential head at design flow (m)
$density_{max}$	=	Maximum density of pumped fluid (kg/m^3) at the pumping conditions (start-up conditions to be considered if sizing but not pre commissioning phase when pumps can be tested with water)
K	=	1.2 for motor driven pump
	=	1.4 for turbine driven pump
A	=	10 000 for pressure unit of measurement in Kg/cm^2

No provision has to be taken for maximum impeller diameter.

In case of two pumps in series, the maximum differential head will be the sum of the maximum differential head of each pump if there is no pressure relief valve between the pumps.

The pump Design Pressure at discharge shall also be applied up to and including the suction block valves for pumps in parallel (for example: spared pumps).

The pump Design Pressure at discharge could also be applied to a vessel emergency isolation valve (EIV) located in the common pump suction line (To be evaluated on a case to case basis, e.g. hydrogen back flow from a reaction section which is a sizing case for relief valve on suction vessel, pump process design pressure shall be applied up to and including the EIV).

7.2.5 Positive Displacement Pump

For positive displacement pumps (rotary or reciprocating type), the discharge design pressure shall be the same as the set pressure (SP) of the external safety valve installed on the discharge piping of each pumps.

The set pressure is determined as follows (based on maximum operating pressure at the discharge)

P operating (kg/cm^2 g)	P Setting (kg/cm^2 g)
≤ 10	$P_{op.}$ + 2.5
10 – 20	1.25 x $P_{op.}$
20 – 50	$P_{op.}$ + 5.0
> 50	1.10 x $P_{op.}$

7.2.6 Design Pressure for Complex Systems

For any piece of equipment in the loop, the design pressure will be equal to the design pressure of the equipment where the safety valve is located plus ΔP, where ΔP is the pressure drop under safety valve discharge conditions, between this item and equipment protected by the safety valve.

For Process systems similar to that of a reactor-recycle gas loop and protected by single safety valve, design pressure shall be estimated based on Compressor settle out conditions as per API STD 521, Appendix B. Minimum design pressure of the separator (PSV set pressure) shall be equal to 1.05 times the settle out pressure.

7.2.7 Tube Rupture for Heat Exchangers

This concerns TEMA and multi tubes heat exchanger and not applicable to double pipe exchangers.

The recommended practice consists in oversetting, if necessary, the design pressure of low pressure side of heat exchangers in order to have a hydraulic test pressure of the low pressure side equal to the design pressure of the high pressure side.

The design pressure of the low pressure side shall be therefore defined as:

$$DP_{low\ pressure\ side} = \frac{10}{13} \times DP_{high\ pressure\ side}$$

This rule, while avoiding the installation of safety valve on exchanger low pressure side for tube the rupture, does not eliminate the possibility of tube rupture itself; the relevant consequences on entire low pressure system should be therefore evaluated case by case. The installation of a relief valve on the downstream piping might still be required especially if vaporization occurs during tube rupture

This rule shall not be applied for column reboilers with the heating medium on the high pressure side considering that an open path exists back to the column.

7.2.8 Steam Containing Equipment

As a general rule, full vacuum conditions should be added to design conditions of a steam containing equipment under normal operation, since vacuum can happen during cooling of such equipment, if it is not connected to atmosphere or equipped with special protection device.

Steam out operation is excluded.

7.3 MATERIALS & CORROSION ALLOWANCE

Metallurgy shall be specified by respective unit designer / Licensor based on process considerations.

7.3.1 Corrosion Allowance

Corrosion allowance shall be provided to meet the minimum design life as specified below based on the corrosion rates for the governing case at the maximum operating temperatures (Note1),

- 30 years for heavy wall pressure vessels (2) (3) (4) and reactors (3) (4)
    - 30 years for non-removable internals
    - 15 years for removable internals
- 20 years for non-heavy pressure wall vessels (3) (4), tanks, pumps, compressors

- 20 years for non-removable internals
- 10 years for removable internals

- 10 years for piping
- 100, 000 hours for heater tubes (5)
- 10 years for S&T heat exchangers tube bundles

Corrosion allowance shall not be lower than the following minimum values (3, 4, 5):

	Carbon Steel / Low-Alloy Steel (6)	**Stainless Steel / High-Alloy Steel (7)**
Wet H_2S service	3.0 mm / 1.5 mm (process) 6.0 mm / 3.0 mm (sour water)	0 mm (if no clad) 3 mm clad thickness
Corrosive process service (excluding Wet H_2S service)	3.0 mm / 1.5 mm	0 mm (if no clad) 3 mm clad thickness
Non corrosive process service	3.0 mm / 1.5 mm (vessels)	0 mm
Water / Steam Condensate	3.0 mm / 1.5 mm (vessels)	0 mm
Steam	3.0 mm / 1.5 mm (vessels)	0 mm

Notes:

(1) Corrosion allowance typically applies to drums, columns, heat exchangers (shell and tubes type, hair pin type) and air coolers. For piping, also refer to project piping classes.

(2) Heavy wall pressure vessels (columns, drums, exchangers, air coolers): above or equal to 50 mm.

(3) No corrosion allowance shall be provided for removable vessel internals in high alloy or stainless steel. The necessary corrosion allowance shall however be specified for internals subject to severe conditions, such as reactor internals.

(4) For internals, each surface side, including welding exposed to the process fluid, shall have a minimum corrosion allowance as follows:
- Welded or non-removable parts: 100 % of the specified value
- Bolted or removable parts (including trays): 50 % of the specified value

(5) For fired heaters, corrosion allowance for process coils(radiation & convection) of carbon steel up to and including 9Cr-1Mo-V tubes shall be 3 mm minimum.
Corrosion allowance shall be minimum 1.5 mm for higher grades of tubes.

(6) Low alloy steel: up to and including 5Cr-½Mo.

(7) High alloy steel: above 5Cr-½Mo up to austenitic stainless steel.

7.3.2 Material Selection

As far as possible, base material shall be Carbon Steel. The following materials are suitable for low temperature service (to avoid embrittlement):

- Carbon Steel (CS) ≥ -29 °C
- Low Temperature Carbon Steel (LTCS) ≥ -45 °C
- Stainless Steel < -45 °C

7.3.3 Special Services Selection

7.3.3.1 Hydrogen Service

Definition

Hydrogen service refers to hydrogen or its gaseous mixtures having a partial pressure of hydrogen equal or higher than 7 kg/cm2(a), or containing 90 volume % hydrogen and higher at pressure level

Requirements (typical)

For minimum base material selection, refer to the Nelson curves, API STANDARD 941 (H_2 corrosion at high temperature). Material selection shall be based on maximum operating temperature (MOT).

For material selection and corrosion rate calculation, refer to Couper-Gorman curves, API RP 939-C, (Guidelines for Avoiding Sulfidation (Sulfidic) Corrosion Failures in Oil Refineries).

Information on Process Documents
Process data sheets shall include a note to clearly identify the hydrogen service. The hydrogen partial pressure of the vapor phase shall also be indicated on the datasheet (ppH_2 in kg/cm^2 abs).

7.3.3.2 Wet H2S Service

Refer to NRL Specification for Carbon Steel Components used in Sour Service in Petroleum Refinery, Document number NR-0ZZZZ-PI-SPE-0025 for applicability of wet H2S service.

Process data sheets shall include a note to identify the Wet H2S Service.

7.3.3.3 Amine Service

Definition

Metallic materials in presence of amine and H_2S in the aqueous phase (free water) may be subject to Sulphide Stress Cracking (SSC) and / or hydrogen induced cracking (HIC).

Amine service is defined as any service that contains 2% or more by weight MEA, DEA, or other amine.

- Amine service shall be specified for items exposed to lean or rich amine service.
- Wet H_2S service shall also be specified for items exposed to lean or rich amine service.

Information on Process Documents

Process data sheets shall include a note indicating amine service and wet H_2S service.
Minimum Material Requirements (typical)

- Base, weld and clad materials shall conform to API STANDARD 581 / NACE MR 0103.
- Carbon steel shall be fully killed and fine grained (KCS).
- Materials shall be stress relieved (post weld heat treatment, PWHT).
- Base, weld and clad materials shall be resistant to hydrogen induced cracking (HIC).

7.3.3.4 Caustic Soda Service

Caustic service is defined as any service that contains 1% or more by weight NaOH or KOH. For caustic soda service, a post weld treatment (PWHT) for stress relief could be required to avoid cracking when using carbon steel.

The caustic service chart in NACE SP 0403 shall be used to determine the need for caustic soda service with PWHT. The selection of caustic service depends on caustic concentration and maximum operating temperature (MOT).

Information on Process Documents

Process data sheets shall include a note to identify the service, i.e. “Caustic Service”.

8. PROCESS DESIGN CRITERIA FOR MAIN EQUIPMENT

8.1 EQUIPMENT DESIGN MARGIN

The overdesign margins on systems and equipment's are considered by respective licensors as per their own standard practice, however the following process design criteria shall be used as minimum requirement to generate the equipment process specification.

Equipment		Overdesign factor (1)
Pumps (2) (3)	**Centrifugal Pumps**	
	Reflux or pump around pumps	20%
	Reflux + Product pumps	20% on reflux + 10% on product
	Reboiler circulation pumps	20%
	Boiler feed water pumps	10%
	Feed pumps	10%
	Product pumps	10%
	Reciprocating Pumps	
	For all pumps	10%
Compressors		
	Recycle compressors in reaction loop (centrifugal)	According to Licensor
	Makeup compressors in reaction loop (volumetric)	According to Licensor
	Other compressors and blowers	10%
Process Turbines (Turboexpanders etc.)		According to Licensor
Fired Heaters	Design duty for process heater shall be:	Minimum design shall be: - 10 % of normal flow for hydraulic margin - And 10% of normal duty or - normal duty + 5 % of exchanger train duty upstream the heater whichever is greater. Or as per Licensor / Designer recommendation
	Reboiler fired heater	As per Licensor requirement or 20% on flow will be specified

Equipment		Overdesign factor (1)
Heat Exchanger	Column reboiler (4)	the highest value between: - 20% on flow and duty - 20% on flow and 10% on duty of feed/bottom exchanger
	Column bottom cooler (4)	the highest value between: - 10% on flows and duty
	Feed preheater or reactor effluent cooler (if not located downstream reactor effluent air cooler) (4)	According to LICENSOR specification
	For feed preheat against products	No margin for thermal rating, 10% on flows for hydraulics on both sides
	For column pump arounds services	20% on duty and on flows of pump arounds (Hot) side. For cold side, margin on flows to be selected in line with service, e.g. 10% if feed or product.
	For product coolers	10% on flow and duty shall be specified
	For column overhead trim condensers	20% on flow and duty shall be specified.
Air Coolers	For column pump arounds	20% on flow and duty shall be specified
	For column overhead air condensers	20% on flow and duty shall be specified. (To be checked against overdesign considered for the reboiler which could lead to higher required margin on the overhead condenser).
	For product cooling	the highest value between: - 10% on flows and duty
	For reactor effluent air cooler	According to LICENSOR specification
Columns	Vapor and liquid load	10% of heat and material balance.
Cooling towers		10%
Package Boilers		Note 5

Notes:

(1) In specific case where, above design margin is different from Licensor value, Licensor's design margin shall prevail.
(2) Continuous recirculation flow shall be added to the net process design flow when installed for minimum flow protection.
(3) Normal and design flow rates shall be identical in such instances as:
   - Intermittent services, sump pumps etc.

(4) To take into account the risk of lower performances of heat recovery systems (e.g. column feed / bottom, reactor feed / effluent) than expected ones, the minimum margins between normal and design shall be specified.
(5) Normal load not to exceed 85% of boiler capacity.

8.2 VESSELS

8.2.1 Types of Vessels

Horizontal vessels shall be preferred for services with high liquid volumes, such as feed surge drum, column feed accumulators, recycle accumulators, liquid separators, etc. Vertical separators shall be preferred when liquid flow-rate is low and vapor phase prevails, or when an accurate level control is required for reduced liquid flow-rates.

8.2.2 Vessel Sizing

Vessels with proprietary internals may be considered, in such a case sizing will be in accordance with licensor / vendor guarantee, otherwise the following guidelines shall be used.

Vessel sizing will be performed as follows:

Vessel length will normally be between 2 – 4 times vessel diameter.

Vertical vessels shall be sized in order to keep vapor velocity sufficiently low and facilitate the separation of the two phases, based on the "critical velocity".

For vessels with internal diameter less than 610 mm, a note shall be added on the process datasheet specifying that a piping element is acceptable. Design adequacy shall be verified by vessel specialist based on applicable pressure vessel design code.

For vessel where entry is required through manhole, the minimum diameter of the vessel is 1200 mm. In some exceptional cases, a vessel diameter between 900 mm and 1200 mm can be specified.

The following level positions shall be used unless specified otherwise by LICENSOR

LSHH (1)(3)	High High Liquid Level (Safety)
HLL	High Liquid Level (Control)
HLA	High Level Alarm (Control)
NLL	Normal Liquid Level
LLA	Low Level Alarm (Control)
LLL	Low Liquid Level (Control)
LSLL (2)(3)	Low Low Liquid Level (Safety)

Notes:

(1) LSHH : Level Safety High High, cut-off (trip) by ESD at high high liquid level.
(2) LSLL : Level Safety Low Low, cut-off (trip) by ESD at low low liquid level.
(3) The trip connections will be independent from other instrument connections.

The following minimum liquid heights shall be used for the level measurement / control:

- Top Tangent Line (TTL) – (LSHH) 300 mm mini. or 20% of diameter whichever is larger
- Top Tangent Line (TTL) – HLL if no LSHH 300 mm mini.
- LSHH-HLL 200 mm mini. (emergency hold-up)
- HLL-LLL 300 mm mini (normal hold-up)
- LLL-LSLL 200 mm mini (emergency hold-up)

The installation of a high-high liquid level alarm shall be considered on the compressor suction separators. If the high-high level alarm is not provided with a shutdown device, an additional 1 minute minimum will be allowed between the high level limit (HLL) and the bottom of the feed inlet nozzle. The same hold-up time value will be provided for vessels with a low/low level alarm.

The recommended hold-up times shall be in accordance with the Table 1.0 attached below

HOLD-UP VOLUMES FOR PROCESS DRUMS (Table 1.0)					
SERVICE	**REFER. SKETCH**	**VESSELS TYPE NORMALLY USED**	**HOLD UP VOLUME (whichever is larger)**		
			LEVEL NORMAL SPAN HL – LL	**LEVEL EMERGENCY SPAN (note 2)**	
				HHL – H L	**LL – LLL**
REFLUX DRUM					
- Liquid product to storage	1	Horizontal	5R or 2P	note 3	1 (P + R)
- Liquid product to fractionator or to Unit without surge drum	2	Horizontal	5R or 10P	note 3	1 (P + R)
- Vapour product	3	Horizontal	5R		1R
LIQUID SURGE DRUM					
- Feed to Unit	4	Horizontal or vertical	10P	2P	2P
- Feed to critical equipment (furnace, column)	5	Horizontal or vertical	5P	2P	2P
VAPOUR / LIQUID SEPARATOR					
- Compressor, suction	-	Vertical	2P (note 4)	2P	-
- Refrigeration compressor, interstage:	-	Vertical	3P (note 4)	2P	-
- Process compressor, interstage:					
- liquid to storage	6	Vertical	5P	2P	
- liquid to fractionator	7	Vertical	5P	2P	2P
- Fuel gas KO drum	-	Vertical	5P (note 5)	-	-
- Steam gas KO drum	-	Vertical	5P (note 5)	-	-
- Steam drum (boiler)	-	Horizontal	(note 6)	-	-
LIQUID / LIQUID SEPARATOR		Horizontal	Time required for separator (note 7)	-	-
COLUMNS BOTTOMS					
- Bottoms to Unit or heat recovery train	8	-	5P	2P	2P
- Bottoms to storage	8	-	2P	2P	2P
- Fired coil reboiler product draw off	9	-	5P	-	2P
- Bottoms to fractionator manual level control	10	-	15P	1P	1P
- Fired coil preheater	11	-	5P to 10P	1P	1P
- Fired coil preheater product draw off	12	-	5 to 10 (F + P)	1 (F + P)	1 (F + R)

HOLD-UP VOLUMES FOR PROCESS DRUMS (Table 1.0)					
SERVICE	REFER. SKETCH	VESSELS TYPE NORMALLY USED	HOLD UP VOLUME (whichever is larger)		
			LEVEL NORMAL SPAN HL – LL	LEVEL EMERGENCY SPAN (note 2)	
				HHL – H L	LL – LLL
- Kettle type reboiler	13	-			
• Bottoms to unit		-	10P	-	2P
• Bottoms to storage		-	5P	-	1P
- Min. hold-up in tar pot	14	-	5 - 10 seconds on P (note 9)	-	-
- Amine absorber	-	-	5P	2P	2P
- Amine stripper	-	-	5P (note 8)	2P	2P

Notes:

(1) P = Liquid product (m³/min.)
R = Reflux (m³/min.)
F = Feed (m³/min.)

(2) Above or below level control span, when HHL or LLL gives shutdown for flooding or emptying.

(3) At least 2R or 1P (whichever is larger) before the flooding.

(4) P is max. liquid flowrate or 10 % of vapor flowrate (on weight basis) whichever is larger.

(5) P is max. liquid flowrate or a liquid volume equivalent to 3 m of feed pipe (whichever is larger) and however not less than 600 mm from BTL to the highest liquid level.

(6) 2F or 1/3 of boiler volume.

(7) or, for each liquid phase, 2P (product to storage) and 15P (product to fractionator) whichever is larger.

(8) If the amine unit includes a surge tank with a hold up volume 15P min., the level normal span can be reduced to 2P; otherwise the level normal span of the stripper bottom is to be designed for 2 – 4 days of amine make-up.

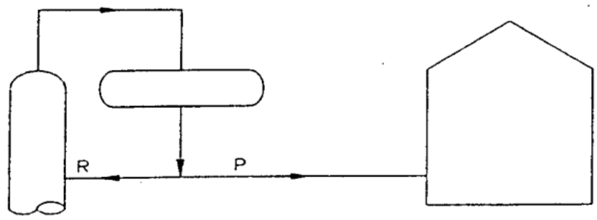
Hold-up volumes for huge plants to be set-up case by case.

(9) Without quench facilities.

REFERENCE SKETCHES

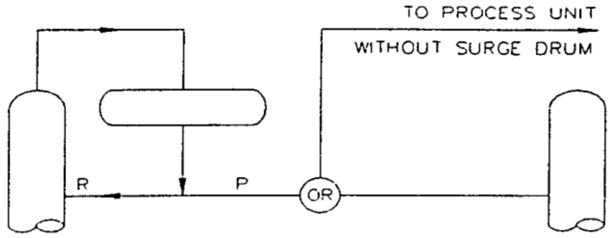
SKETCH N. 1

LIQUID PRODUCT TO STORAGE



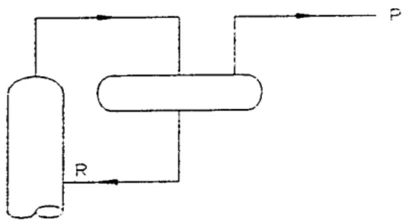
SKETCH N. 2

LIQUID PRODUCT TO FRACTIONATOR OR TO UNIT WITHOUT SURGE DRUM



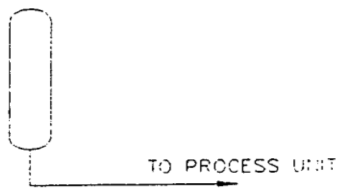
SKETCH N. 3

VAPOR PRODUCT



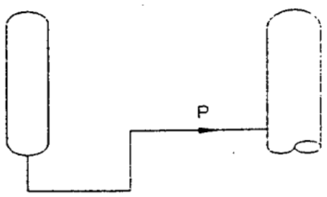
SKETCH N. 4

FEED TO UNIT



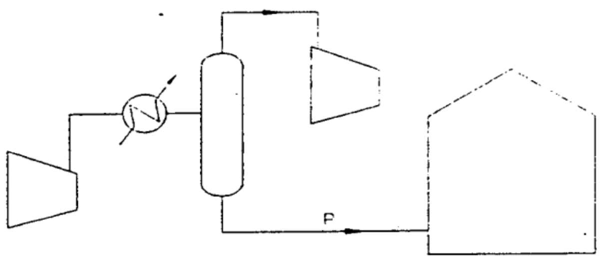
SKETCH N. 5

FEED TO CRITICAL EQUIPMENT (FURNACE, COLUMN)



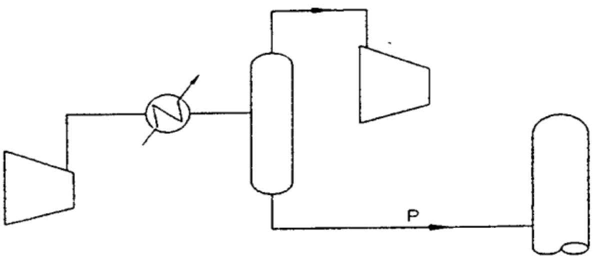
SKETCH N. 6

PROCESS COMPRESSOR INTERSTAGE LIQUID TO STORAGE



SKETCH N. 7

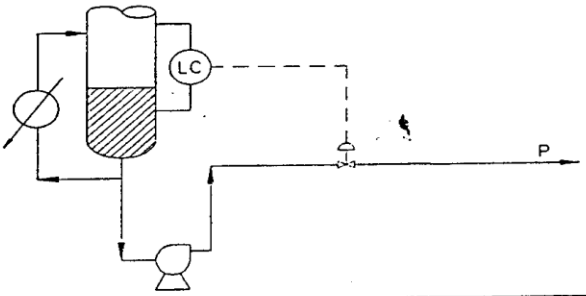
PROCESS COMPRESSOR INTERSTAGE : LIQUID TO FRACTIONATOR



REFERENCE SKETCHES

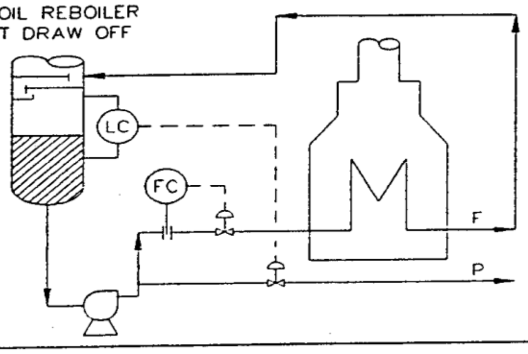
SKETCH N. 8 COLUMN BOTTOMS

TO UNIT OR TO STORAGE



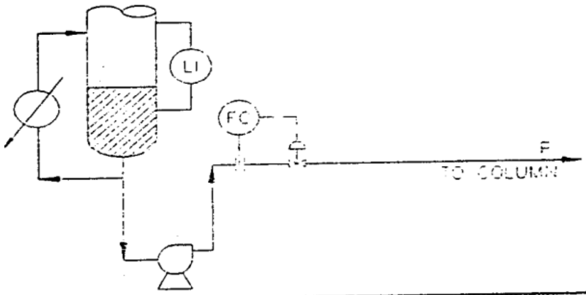
SKETCH N. 9 COLUMN BOTTOMS

FIRED COIL REBOILER
PRODUCT DRAW OFF



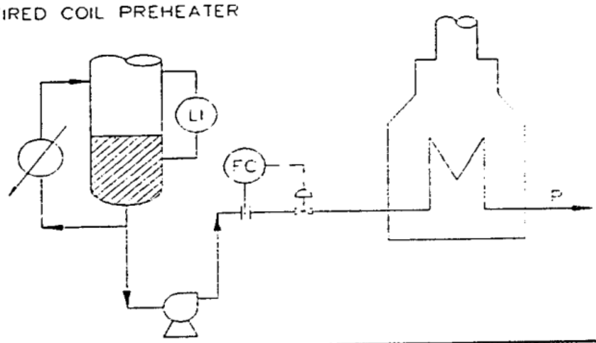
SKETCH N. 10 COLUMN BOTTOMS

TO FRACTIONATOR
MANUAL LEVEL CONTROL



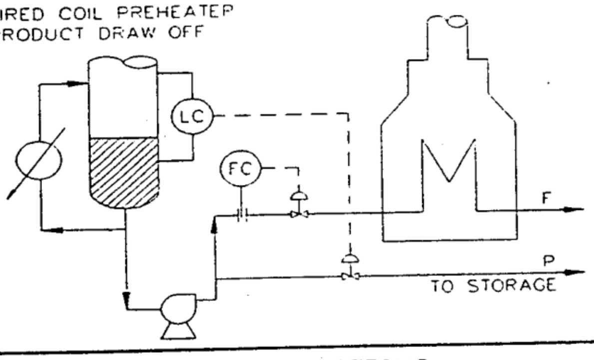
SKETCH N. 11 COLUMN BOTTOMS

FIRED COIL PREHEATER



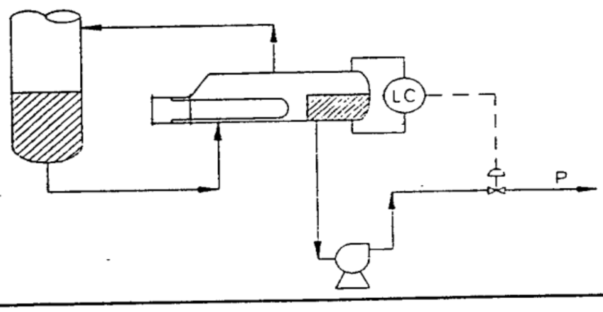
SKETCH N. 12 COLUMN BOTTOMS

FIRED COIL PREHEATER
PRODUCT DRAW OFF



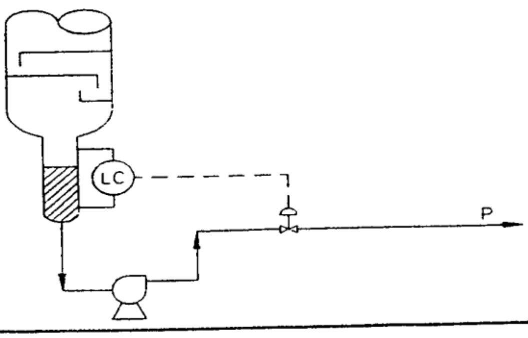
SKETCH N. 13 COLUMN BOTTOMS

KETTLE-TYPE REBOILER :
TO UNIT OR TO STORAGE



SKETCH N. 14 COLUMN BOTTOMS

MIN. HOLDUP IN TAR POT



8.2.3 Equipment Nozzles

Nozzle Connections

Unless otherwise dictated by Process or Licensor requirements, the vessel's connections shall be minimized by installing Relief valves, temperature and pressure measurements connections on piping where possible instead of on vessel.

Minimum Nozzle Size: The minimum recommended nozzle connection size shall be

- For flanged connections attached to unclad vessel shall be 2"
- For flanged connections attached to internal lined / clad shall be 3"

Refer to Engineering design basis for Instrumentation document number TP-1ZZZA-IC-BOD-0001 for instrument nozzle size and connection details.

8.2.4 Manholes Handholes & Flanged Heads

The minimum size of manholes shall be

- 24" NB for vessel diameter ≥ 1 200 mm
- 20" NB for vessel diameter between 1 000 mm to 1 199 mm
- 18" NB for vessel diameter between 900 mm to 999 mm (to be decided on case to case basis)

Flanged heads shall be specified for diameter smaller than 900 mm when access for vessel / column internals is required and where no access is required a hand hole will be provided with minimum of 8" NB unless a suitable nozzle for other service exists.

For horizontal vessels having a straight length between heads welds greater than 6 000 mm, one manhole shall be provided on each opposite side (on heads or shell). For shorter horizontal vessels, at least one manhole shall be provided plus a separate blanked-off ventilation nozzle on the opposite side of the manhole (or to the inspection hole) such as to provide effective ventilation during inspections, refer to Section 7.2.6 Ventilation Nozzle.

For vessels with diameter lower than 900 mm which cannot be physically entered, manholes shall be replaced by handholes (one or two) minimum of 8" size.

8.2.5 Drains Vents & Utility Connections

Drains, vents and utility connections for vessels shall be sized as follows:

Volume of the vessel (m^3)	Vent size	Drain size	Utility connection size
$15 < V \leq 75$	2"	2"	2"
$75 < V \leq 220$	3"	3"	2"
$220 < V \leq 420$	4"	4"	3"
$V > 420$	6"	4"	4"

Vessels with internal baffling require special consideration: one drain on each side of the baffle.

Utility connections shall be hard-piped for nozzle sizes of 3" and above. All hard-piped utility connections to be provided with block and bleed arrangement with spectacle blind, swing elbow to be considered for specific case.

8.2.6 Ventilation Nozzle

If entry in a horizontal or vertical vessel (with length of 3 000 mm or above) equipped with only 1 manhole is required, a blanked-off ventilation nozzle shall be added directly to the vessel in addition to the vent. The vent with blanked-off valve which is typically used in steam-out operations can directly mounted on the ventilation nozzle.

For horizontal vessels, 3 000 mm or over in tangent length, provide blanked-off ventilation nozzle on the top of the vessel near the end and opposite to the manhole.

For vertical vessels, 3 000 mm or over in tangent length, provide blanked-off ventilation nozzle on the top head unless two (or more) manholes are specified with at least one at the top and one at the bottom of the vessel.

Ventilation nozzles shall be sized as follows:

Vessel Length (mm)	Ventilation Nozzle Size
TL to TL < 3 000	-
3 000 ≤ TL to TL ≥ 4 500	4”
4 500 ≤ TL to TL ≥ 7 500	6”
TL to TL > 7 500	8”

8.2.7 Vortex Breakers / Mist Eliminators

Liquid discharge nozzles shall be provided with vortex breakers in the following cases,

- Connection to pump suction
- Connection to reboiler inlet
- Liquid draw-off from trays
- Connection to control valve

For fouling services, the vortex breaker shall be installed at a minimum of 150 mm above the vessel bottom. For drums equipped with a boot in water / hydrocarbon separation service, the vortex breaker shall be installed at a minimum of 150 mm above vessel bottom.

Mist eliminator shall be installed whenever liquid droplets must be removed from vapor to the maximum extent.

8.3 COLUMNS & TRAYS

For general requirements applicable to all vessels refer also to Section 7.2 Vessels.

In general the following criteria is to be followed for number of pass trays, unless specified otherwise by Licensor

- Single pass, 2 pass and 4 pass trays can be utilized appropriately.
- Utilization of 3 pass trays are generally not preferred.

8.3.1 Types of Trays & Tray Sizing

Trays shall be numbered from bottom to top of the column.

In general, valve trays shall be used, sieve trays may be used in fouling service. Foaming factor shall be specified for columns in foaming service.

The column shall be sized based on the design vapor / liquid loads, foaming factor and the following maximum flooding factors calculated at design flows:

- 70 % for columns with diameter below 900 mm
- 77 % for vacuum columns
- 80 % for other columns

For columns of diameter ≤ 900 mm, trays shall be cartridge type trays.

8.3.2 Tray Spacing

Minimum tray spacing shall be followed unless specified otherwise by Licensor (only for conventional trays):

- 450 mm for column diameter ≤ 2 000 mm
- 600 mm for column diameter > 2 000 mm
- 1 000 mm for installation of a manhole (24")

8.3.3 Tray Loading Margin & Flexibility

Tray hydraulic calculations and confirmation of column diameter shall be done by the tray vendor. Required tray flexibility shall be 50 - 110 % of guaranteed cases, unless otherwise specified to reach operating conditions flexibility.

8.3.4 Tray Material

Tray material is specified, as a minimum, in SS410S without corrosion allowance, however for licensed units licensor specific recommendation shall be followed

8.3.5 Column Internals / Connections / Hold-ups & Height

Thermowells shall usually be located at feed trays, side stream trays and trays for reboiler control and shall be installed in the vapor phase 200 mm (8”) below upper tray.

Manholes shall be provided for the installation and access of column internals. The arrangement shall be determined by the pressure vessel specialist (e.g. orientation and elevation).

The minimum size of manholes shall be 24". Manhole sizes of 20" or 18" may be used on case by case basis for vessel with diameter less than 1 200 mm. For vessels with internal diameter less than 1 200 mm, flanged heads can also be specified (without installation of manhole) if access for the vessel internals is required.

For packed columns, each packed bed will have a manhole each above and below the bed. For packed beds with random packing, hand holes shall be provided just above the packing support plates, for removal of packing. These shall be two in number and opposite to each other for large diameter columns (diameter 1000 mm and above).

To facilitate access to the column internals, minimum number of manholes shall be positioned as follows:

- Top near the head
- Feed / return tray (for easy access to distributors)
- Below draw-off tray
- Flash zone
- Bottom near the head
- At any tray where removable internals are located.

In addition, depending on the service, the following table applies:

Service	Additional Manhole location
Clean	1 manhole every 15 trays
Fouling	1 manhole every 10 trays

Hold-up times for sizing column bottoms shall be in accordance with the Table 1.0 under section 7.2.2 vessel sizing.

8.4 AIR COOLERS & HEAT EXCHANGERS

8.4.1 Air Coolers.

- A maximum of air coolers should be installed Yes…………
- Type of control, if any, to be considered:
  - Automatic variable pitch No…………
  - Variable speed motors Yes…………
  - Shutters No…………
  - Air recirculation No…………
- Tubes with or without fins acceptable in one or two tube rows in a tube bundle Yes…………

8.4.2 Selection of Air Coolers

- Air cooled heat exchange is to be maximized except for specific intermittent services or small cooling duties where air-cooled exchanger is not justified.
- The lowest process stream outlet temperature for air cooled heat exchange is:
  - Air cooled exchanger not followed by water cooling : 50 °C
  - Air-cooled exchanger followed by water cooling : 60 °C
  - Dry bulb temperature for cooler sizing : 35 °C

  However, to avoid small trim cooler or air cooler, these guidelines can be relaxed.
- Forced draft fans are preferred for air cooled heat exchangers. However, for the low approach temperature less than 12°C, induced draft fan shall be used in consultation with owner and M-

PMC. If induced draft fan to be used, Licensor has to refer to owner / M-PMC on case to case basis.

- Fan blades in FRP / Aluminium are preferred.
- Air cooler air side fouling resistance = 0.0001 hr.m2.C/kcal.

- Preferred location / sizes for Air-cooled exchangers: Air coolers may be located on pipe-rack or platform structures.
- Air-cooled exchanger tube length to be used depends upon pipe-rack width / equipment layout.
  - Typically tube length should be pipe rack width + 0.5 m.
  - Preferred tube length is ______ m (to be confirmed later).
  - Air cooled exchanger bundle width : 3.2 m. (4.0 m max. for single Bundle)
  - Limits on fan motor power with (HTD) belt drive : 45 kW
  - Fan Motors over 45 kW shall use Gear drives.
- Requirements of winterizing or for highly congealing service air coolers, following option will be considered:
  - Steam coils.
- Avoid air coolers in highly congealing service.

Preferred Tube Dimensions

Tube Metallurgy	CS / LAS	HAS / SS
Tube diameter	25.4 mm OD	25.4 mm OD
Tube thickness	2.77 mm	2.11 mm
Fin height	15.875 mm / 12.5 mm	15.875 mm / 12.5 mm
Type of Fin	Al. `G’ type *	Al. `G’ type *
Max. No. of fins	433 per m	433 per m

* G – Grooved (embedded) fins

For design temperatures > 400°C, CS welded fins shall be used. Fin density for CS fins shall be 236 per m in this case.

8.4.3 Type of Headers

Plug header shall be provided if any of the following conditions exist: Otherwise cover header are to be provided.

- Hydrogen partial pressure ≥ 7.0 kg/cm² (a)
- Clean service with fouling < 0.0004 Hr. m2 °C/kcal
- High pressure over 30 kg/cm² (g)

If inter-pass temperature difference exceeds 110°C or if inlet temperature exceeds 180°C, split headers shall be used. In addition to this Licensor requirements shall prevail in selection of type of headers. Minimum slope of 10 mm / meter shall be provided for single pass condensing exchangers, towards the outlet header. Last pass of multi-pass condenser shall have a slope of 10 mm / meter, towards the outlet header.

8.4.4 Grouping of Air Coolers

Large air-cooled exchangers shall be divided into sections and fans sized and provided in each section. Air-coolers, which are smaller than section size, can usually be grouped into one section and provided with a common set of fans (for similar service), unless the designer has an over- ruling consideration. However, air coolers provided with control arrangement cannot be combined together.

Inlet and outlet piping for air coolers with multiple bundles / bays shall be designed to ensure uniform process fluid distribution and avoid channeling.

8.4.5 Fin Fan Control Scheme

When fluid outlet temperature requires to be controlled (whether automatic or regulated manually, say, for seasonal climatic variation) or when stipulated to conserve power, variable speed drives are to be considered.

Variable frequency drives (VFD) are the preferred means for speed control of air cooler fans where airside flow control is required minimum 50% of total fans shall be provided with VFD.

8.5 HEAT EXCHANGERS

8.5.1 Plate Heat Exchangers

Plate type heat exchanger can be installed in place of shell and tube heat exchanger as deemed fit for best heat recovery by the unit designer / Licensor. Use of these exchangers should not have effect on unit reliability. Adequate references should be available for similar service, fluid quality and trouble free operation.

8.5.2 Shell & Tube Exchangers

Fouling Factors

Fouling factors for process streams: Fouling factors for process streams are decided by Licensor / designer based on past experience and good design practices. However, specific requirements are addressed in unit design basis document.

Fouling factors for Utility streams: The following default fouling factors for utility streams are to be considered.

Cooling Water:	0.0004 hr-m2-ºC/kcal
Steam:	0.0001 hr-m2-ºC/kcal
DM Water:	0.0001 hr-m2-ºC/kcal
Tempered Water:	0.0002 hr-m2-ºC/kcal
Condensate:	0.0001 hr-m2-ºC/kcal
Air:	0.0001 hr-m2-ºC/kcal
For Others:	As per TEMA

Tube-Side Velocities

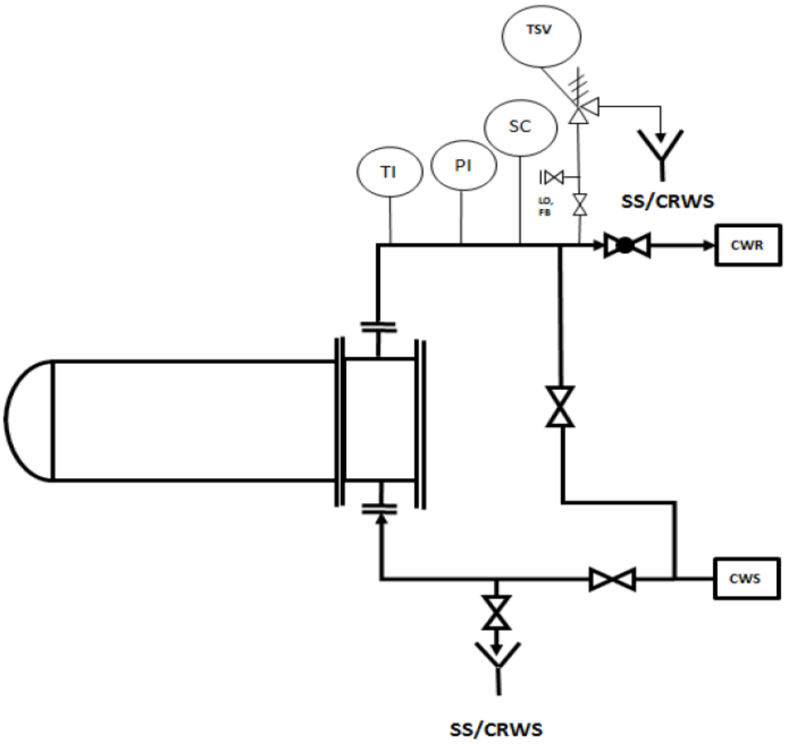
It is a good practice to recommend specific tube side velocities for better on-stream factors of heat exchangers in fouling services. Cooling water flow rates are normally not expected to be turned down during operation and can be designed for a velocity constraint applied only for normal flow.

Minimum tube side velocity for cooling water shall be 1.0 m/sec at normal flow. Process fluid velocities are to be specified by licensor / designer considering the nature of fluid, e.g. viscous, fouling, slurries etc. Cooling water heat exchangers to be designed with maximum allowable pressure drop of 0.7 kg/cm2.
For ease of cleaning, cooling water to be kept on the Tube side.

Minimum tube side velocities for specific services such as slurry oil, MCB exchangers shall be followed as per licensor requirement.

Typical Cooling Water Piping

All cooling water consumers connected to recirculation cooling water system are to be provided with back flush arrangement. The Typical scheme is provided below, the detailed requirement is provided in legend and typical PIDs.



Note:

(1) Back flush lines to be provided with same size as main cooling water line when main line size is ≤ 6 inch. One size lower to be provided for main line size up to 10 inch. Two size lower to be provided for main line size > 10 inch.
(2) For exchangers at an elevation, vacuum protector to be provided in cooling water lines.

8.5.3 Selection of Shell & Tube Heat Exchangers

Floating Head type exchanger is preferred construction in refinery service. The use of fixed tube-sheet heat exchangers shall be minimized to the extent possible. These can be used in limited services such as vertical thermosyphon reboilers where steam is on shell side. Steam-out conditions shall be specified for fixed tube-sheet heat exchangers, considering either shell-side or tube-side being steamed out at one time. Expansion bellows are not preferred on fixed tube sheet heat exchangers.

Double pipe or multi-tube heat exchanger should not be used except in the event of infeasible shell-and-tube heat exchanger design.

The exchangers can be stacked considering the size of exchangers, equipment layout and approach for operation and maintenance.

Maximum number of shells in a stack

- For shells with diameter <500 mm : 3 (if individual shell isolation and bypass facility not provided).
- For shells with diameter >500 mm : 2 (if individual shell isolation and bypass facility not provided).
- Exchangers not to be stacked, when individual shell isolation and bypasses are provided.

General Criteria for Selection of TEMA Type

Shell side fouling resistance (hr-m^2-^{o}C/kcal)	Tube side fouling resistance (hr-m^2-^{o}C/kcal)	TEMA type (preferred)
>0.0002	>0.0002	Floating head
≤0.0002	>0.0002	Fixed tube sheet
>0.0002	≤0.0002	U tube bundle
≤0.0002	≤0.0002	Fixed tube sheet / U tube bundle

Or as specified by licensor

All shell and tube heat exchangers in a refinery application shall follow TEMA-R unless required otherwise. However, for coolers (oil coolers) in the packaged equipment's will follow TEMA-C. Process surface condensers (e.g., vacuum producing package of CDU/VDU unit) in the package scope will be TEMA-R.

Preferred exchanger type for refinery services shall be horizontal, single pass shell, floating head tube bundle (TEMA type 'S') arranged with two or more tube passes per shell. Specific type recommended by Licensor to be complied with.
Tube pitch shall be square or rotated square for fouling services and triangular pitch for clean services.

The fouling factors ≥ 0.0004 hr-m2- ^{0}C/kcal is defined as dirty service.

General Criteria for Specifying U-tube Construction is:

- Services where steam is condensed or generated on tube side; tube pitch shall be triangular if shell side fluid is clean and square / rotated square if shell side fluid is fouling.
- TEMA type B stationary head: hydrogen services or when steam or other clean fluid is on tube side, not requiring cleaning.
- TEMA type A stationary head when tube side fluid is cooling water or similar process fluids, requiring cleaning.
- For U tube bundles, consideration shall be given for hydro testing of tubes for IBR/normal testing after preventive maintenance. If necessary, dummy shell shall be designed for testing for maintenance purpose.
- Other constraints: TEMA type B or C stationary heads shall not be used when design pressure is higher than 140 kg/cm2 g. Tube sheet shall not be full diameter extended type when TEMA type B stationary head is used.

Cathodic Protection Requirement for Cooling Water Exchanger

- Requirement of cathodic protection for cooling water side of exchangers in case of dissimilar metallurgy only. Sacrificial anode, if required, shall be sized and positioned in such a way that flow restriction is not there.

Preferred sizes for shell & tube Exchangers with Removable Bundles

- Preferred Tube lengths : 2.438 m, 3.048 m, 3.658 m, 4.877 m, 6.096 m, 7.315 m and 9.144 m.
  Max. tube length - 6.096 m, for exceptional cases tube length up to 9.144 m shall be used.
  Maximum tube length shall be 9.144 m for fixed tube sheet exchangers.
- Max. tube bundle diameter : 1.5 m for 6.096 m tube length, For exceptional cases tube bundles diameter may exceed by 100 mm.
- Max. tube bundle weight : Approximately 25 tons and for high pressure exchanger, bundle weight can go higher.

Preferred tube dimensions

Type	CS & Low alloy	High Alloy & SS
Tube diameter	25.4 mm /19.05 mm	25.4 mm /19.05 mm
Tube thickness	2.77 mm / 2.11 mm (min.)	2.11 mm / 1.65 mm (avg.)
Tube thickness (IBR Service)	2.77 mm / 2.11 mm (min.)	2.11 mm / 2.11 mm (min.)

Any change in the above preferred dimensions, specific approval is needed.

For Licensed units, deviations on tube size and thickness are acceptable as Licensor requirement is to be followed. However, for IBR service, the tube thickness specified in the table shall be met as a minimum requirement.

For dirty service, tube OD >= 25.4mm shall be used.

Note: Higher tube OD to be used in case of pressure constraints.

- Low fin tubes are not allowed.

- Longitudinal fin tubes are not allowed.
- Use of tube inserts is allowed for viscous services with approval from Owner / M-PMC.

8.5.4 Additional Instructions

For shell-and-tube heat exchangers, the low pressure (LP) side of exchanger including upstream and downstream system equipment shall be preferably specified with a design pressure at least equal to 10/13 of high pressure (HP) side design pressure, in order to avoid having to install a pressure relief device on the LP side.

The 10/13 criteria shall necessarily be adhered to when:

- LP fluid is on the tube side.
- Relief discharge cannot be connected to flare header owing to nature of fluid.
- Relief discharge is two-phase and cannot be connected conveniently to a low-pressure destination with free-draining piping.
- Liquid relief cannot be connected to a closed blowdown system.

When the 10/13 criteria call for an increase by less than a factor of 13/10 of the mechanical design pressure of the LP side as would be calculated from normal estimation procedures. However, based on process considerations and designers own practice, heat exchanger can be designed without upgrading design pressure by providing tube rupture pressure relief valve after getting consent on case to case basis from Owner / M-PMC.

For feed / effluent exchanger, the tube sheet and tubes in high pressure service need not be designed for full design pressure of the shell and / or channel provided these components can never experience these conditions. The same should be designed for maximum differential pressure expected. Designer must consider start-up, shutdown and emergency depressurizing while deciding maximum differential pressure.

In cases, where shell side pressure is very low compared to tube side, for subsequent testing of tube bundle, additional facility like spare shell for testing shall be considered.

- Mechanical / Chemical cleaning connections shall be provided as applicable
- Impingement protection (plate/rods) to be provided on tube bundle for inlet side of shell as per TEMA criteria.
- Unless otherwise specified by the licensor standards, for shell ID up to DN400 (CS shell), nominal pipe shall be considered.

8.6 PUMP

8.6.1 Pump Sizing Criteria

Pump design flowrates shall include a minimum of 10 % margin on max. operating flowrate unless in the following cases:

- intermittent service
- recirculation

for which the design flowrate will be equal to operating flowrate.

Note: When a permanent recirculation flow for mini flow protection is installed, extra flow must be added to the net process flow.

Pump differential head indicated in the specification shall be calculated at design flow-rate.

8.6.2 Pump Sparing

Process pump in continuous service will, in general, be provided with installed spare. Intermittent pump services such as chemical unloading, batch make-up, etc., shall not be provided with installed spare, however one complete spare pump shall be supplied as warehouse spare. These shall be decided on a case-to-case basis.

For pumps in similar service (such as HC drain pumps), a common warehouse pump shall be considered.
Where more than one operating pump is found necessary for a service, or when defined by owner, pump capacity and spare shall be as follows:

Number of operating pumps	Rated capacity per pump	Installed spare pumps
1	100% of total rated flow	1
2	50% of total rated flow	1
3	33% of total rated flow	1
4	25% of total rated flow	2

As an operational requirement, Boiler Feed Water (BFW) service shall be provided with minimum of two operating pumps with different power source shall be considered.

8.6.3 Seals

Selection of Mechanical seals shall be as per API 682 guidelines and OISD STD125 latest edition.

8.6.4 General Recommendations

Positive displacement pumps shall normally be provided with a relief valve.

Reciprocating pumps shall have pulsation dampeners on the suction and discharge side whenever liquid pulsation is detrimental to process operation.

Auto-start for pumps shall be considered on a case by case basis based on criticality or as per Licensor’s recommendation.

End of curve operation will be specified on pump data sheet for the following cases:

- pumps in auto start service (remote control, sequence activation, etc.).
- pumps without control valve at discharge.
- pumps in parallel operation.
- Tank farm pumps & Utility pumps (case to case as required).

8.6.5 Automatic Stop by LSLL

An automatic stop (trip on LSLL) shall be provided in the Emergency shutdown System (ESD) in case of low low liquid level for the following cases:

- Presence of light hydrocarbons of liquid butane or a more volatile liquid under normal operating conditions (e.g. LPG)
- Liquids above their auto-ignition temperature (AIT)

The ESD interlock prevents the pump from running dry which could cause mechanical damage to the pump and to the mechanical seals followed by the release of a dangerous fluid to the atmosphere.

8.6.6 Minimum Flow Protection & Control

A minimum flow recirculation with flow control shall be implemented for centrifugal pump for the following cases (but not limited to):

- Multistage pumps with differential pressure higher than or equal to 35 kg/cm2g.
- Pumps with discharge control valve operated by suction side vessel level with (LIC / FIC Cascade).
- Low capacity pumps (< 10m3/hr) on case to case basis.
- Large pumps with driver power higher than 160 kW.
- For process reasons (turndown), flow rate lower than or equal to 30% of maximum flowrate.
- For pump with capacity less than 100 m3/hr or driver power less than 18 kW, the MFB usually will be a spillback line with constant bypass flow through a restriction orifice and a globe valve. A separate min. flow control for each pump (operating as well as standby) shall be evaluated for high pressure, multistage pumps.
- For pumps with capacity more than 100m3/hr or driver power more than 18 kW, except recirculating service pumps, shall be provided with an automatic min. flow control with a protective minimum flow control provided to save the pumping energy.
- For low capacity pumps if min. flow control will be through a restriction orifice, minimum flow shall be added to maximum pump flow for operating case to specify the rated flow.

8.7 COMPRESSORS

8.7.1 Centrifugal Compressors

Unless there are specific Licensor requirements, the design capacity of compressors shall include a 10 % minimum margin over the operating flow rate for each centrifugal compressor except recycle compressors in reaction loops.
Compressors shall be designed (in terms of capacity, head, etc.) considering all possible operating cases (e.g. start-of-run, end-of-run etc.) The driver rated power shall cover all possible operating cases (e.g. SOR, EOR, Star-up etc.)

8.7.2 Reciprocating Compressors

Flow-rate control on process compressors shall have the following points of reference: 0 %, 25%, 50 %, 75% & 100 %, or else in accordance with process requirements and number of cylinders per stage. Step less control shall be considered on case by case basis.

8.7.3 Sparing Philosophy

- **Centrifugal Compressor:** Normally no spare compressor shall be considered. Spare rotor shall be considered.
- **Screw Compressor:** Normally no spare compressor shall be considered. Spare rotor shall be considered. For screw compressors in refrigeration service, 100% spare shall be considered.
- **Reciprocating Compressor:** 100% stand-by to be provided for continuous operating reciprocating compressor. For Specific intermittent service, no spare compressor is required.

8.8 DRIVERS

8.8.1 Steam Turbine Drives

Steam turbine drives shall be specified in extremely critical services where even short-term failure of a drive can result in a shutdown from where an operational recovery is difficult, time-consuming or has a large economic penalty, such as irreversible catalyst poisoning.

Steam turbine drives shall also be specified for the following drives that, among other considerations, shall ensure that a power failure does not automatically lead to a steam failure:

Options	Yes/No
Cogeneration / Steam generation plant BFW pumps	Yes
Compressor lube oil & seal oil pumps case to case basis	Yes
Emergency Evacuation pumps	Yes
All utility drives which are required to operate during power failure.	Yes
Any specific drive which can help in mitigation of large flare load during Power failure scenario.	Yes

8.8.2 Preferred Type of Steam Turbine Drive

Back pressure turbine is preferred type when there is a large use of MP or LP steam, a back-pressure steam turbine to drive a pump / compressor shall be evaluated in the unit to minimize let down through pressure reducing station (PRDS) from HP steam level to MP / LP steam level.

Steam condensing pressure at turbine exhaust shall be considered as 0.12 kg/cm² (a) for the purpose of basic engineering specifications. Only water cooling shall be considered for exhaust condensation. Cleanliness factor of 0.6 shall be considered for surface condensers.

In case back pressure turbine is considered, HP to MP turbines are preferred.

8.8.3 Critical Service Rotating Equipment

Motor drives are preferred for pumps and Reciprocating compressors. In the event of general power failure, emergency power supply may sometimes be required to aid in safe shutdowns or to minimize impact of normal power supply failure. Analysis of such emergency power supplies usually lead to a requirement of continuously running a diesel generator or to ensure backup motive steam drives in power plants.

It is recommended that requirement of emergency power be minimized through strategies such as steam turbine drives for standby pumps, steam turbine drive for some cooling water pumps, etc. Emergency power drive requirement can be also specified / identified to reduce overall flare load during General Power failure depending on feasibility.

Unit designer / licensor shall specify the requirement of emergency power, if any, within ISBL based on their experience.

In general,

- If emergency oil pump is provided for compressors, the same to be provided with emergency power backup in case of AC motor drive.
- Emergency power back up shall be provided for CBD pumps (case to case basis) and Flare KOD pumps.
- Nominal requirements of emergency power such as for local panels, frame heating of compressors, etc., can be checked and implemented wherever required

Critical lighting only will have DC supply. The emergency lighting will have AC supply and fed from emergency power source (DG).

8.8.4 Reacceleration Feature

Designer/ Licensor may specify reacceleration feature in some critical pump motor drives to ensure faster start-up or to minimize plant down time in case of process disturbances due to momentary voltage dips.

Requirements	Yes/NO
No reacceleration feature is desired	No
Reacceleration to be specified for identified motors to cover brief interruptions up to 5 seconds in normal power supply. Such pumps to be identified based on designer / licensors own experience.	Yes
a) Re-acceleration feature is acceptable for critical pump motors of 415 V rating (LT) and auxiliary drives of HT motors such as lube oil pumps etc. b) For HT motors, reacceleration feature shall be provided wherever required for process consideration as per licensor / designer	Yes

8.9 HEAT RECOVERY

Heat integration is to be done in the Process units as far as possible. Designer / Licensor of a particular unit shall indicate the following at the outset of design activities,

- Energy shall be preferentially recovered into process streams. Steam generation MP steam (preferred), LP Steam shall be considered thereafter to recover excess available energy. Steam generation levels shall be chosen to preferably match the corresponding steam level demand within unit. Run down stream temperature to Cooling water exchanger or air cooler to be minimized.
- For reboilers, heat input using Process stream shall be preferred option over steam. To be decided by unit designer / licensor.
- Option of the heat integration within the same BLOCK to be explored during the Basic design stage, taking in to consideration operational flexibility / reliability of such integration.
- Low-level energy recoverable for external consumption, say, for Boiler Feed Water/DM water preheats which can be used in other Units. This requirement shall be chosen to match the total complex requirement.
- Option of Liquid ring vacuum pump in place of last stage of ejector shall be explored (for energy / operating cost).
- Maximize heat recovery / Steam generation from heater convection section.
- Deployment of PHE for increasing low level heat recovery.
- Application of PRTs wherever possible
- Application of VFDs in Pumps, Air coolers, Steep less control in compressors.

8.10 HEATERS

Unless there are specific Technology Licensor requirements, fired heaters should be designed with the following criteria:

- Operating Range : 30% - 100%

Unless otherwise specified by the Technology Licensor, the following criteria shall be used for fired heaters overdesign, whichever is greater :

Design duty for process heater shall be:

- 110 % of normal duty (or)
- Normal duty + 5 % of exchanger train duty upstream the heater
- For Licensed Units design, licensor specifications shall be governing.

8.10.1 Type of Fuel for Firing

Heaters shall be of Gas fired with Fuel Gas / RLNG.

8.10.2 Heater Efficiency

Achievable heater efficiency depends on service, process temperatures, furnace heat duty and quality of fuel. Highest target efficiencies shall be pursued by a unit designer, as found economically justified. Air preheating is preferred to steam generation for heat recovery for better furnaces efficiency.

Fuel efficiency calculation shall be based on the Low Heating Value (LHV) of the fuel. Efficiency calculation shall include a minimum radiation loss of 1.5% in case there is no air preheat system and minimum 2.5% in case there is an air preheat system.

Desired minimum heater efficiency based on low heating value of fuel: and ____Note-1____% excess air for gas firing : 92% (min)
In case of minimum guaranteed efficiency value, individual heater Process datasheet shall be followed.

Note-1:

- For Natural Draft
  - Fuel Oil 25%
  - Fuel Gas 20%

- For Force Draft
  - Fuel Oil 20%
  - Fuel Gas 15%

However, hydraulic design of flue gas duct and stack to be suitable for 40% excess air, For Licensed units / design, licensor specifications shall be governing.

8.10.3 System for Waste Heat Recovery from Flue Gas

- Steam generation ☐

Pressure level requirement : HP / MP / LP Steam

- Air preheat ☒

Rotary regenerative ☐

Stationary tubular ☒

Circulating fluid ☐

Other ☐

Absolute Heat Duty (MM kcal/h)	Air Preheater Design
<7	None
≥7<15	On board
≥15<20	On board / Out board
≥20	Out board

APH type: Provide a flue gas air preheater of the recuperative (stationery) type. Cast iron type air preheater shall be supplied. Option of using Welded Plate type heater (preferably) to be consider.

- The fuel gas temperature leaving the flue gas/combustion air preheater shall be 40 °C (75 ˚F) above the maximum calculated acid dew point temperature.
- The minimum metal temperature shall be at least 14°C above acid dew point of flue gas.
- For Licensed Units design, licensor specifications shall be governing.

Steam generation from flue gases to be minimized by capturing the heat into combustion air to the maximum extent possible. Subsequently LP steam or MP steam generation to be considered.

APH wash effluents are collected in underground pit, neutralized by caustic and plant air supplied to the pit to ensure mixing of caustic and effluent. Neutralized effluent is pumped out by steam ejector to OWS.

8.10.4 Burner Minimum Turndown based on Design case

- 5:1 for gas firing
- In case of limitation of turndown, few burners may be kept off

8.10.5 Indicate any other process requirements relating to fired heaters

Maximum allowable average heat flux in the radiant section shall be follows:

Configuration	Center-Center Nominal spacing	Maximum Allowable Average Heat Flux, W/m2 (Btu/h-ft2)
Single Feed	2 c-c	31 550 (10 000)
	3 c-c	37 850 (12 000)
Double Feed	All	45 750 (14 500)

As per Licensor PDS and specification. Pollution monitoring nozzle required as per state / central pollution control board.

8.10.6 Control of FD/ID Fans

Recommended combustion air configuration for forced-draft (FD) heaters is considering each FD fan sized for 100% capacity but with both running normally at 50% capacity. Failure of one FD fan shall cause the other fan to switch over to 100% load. Total fan failure shall result in heater shutdown and no stipulations shall be made for automatic switchover to natural draft operation.

Cold air bypass to be provided and heater specifications shall allow operation at 100% normal duty with APH bypassed.

Spare induced-draft (ID) fans shall not be provided. Failure of ID fans shall cause APH bypass to open to divert flue gas to the stack. Detail heater design shall have provision for coil purging (as applicable), snuffing steam and water washing of FD/ID fan impellers and out-board air preheaters.

8.10.7 Capacity Control

- **FD Fans** **Tick mark**
  - Variable Frequency Drive ☒
  - Variable Frequency drive and inlet box damper ☐
  - Hydraulic Coupling / Variable Speed Drive ☐
  - Inlet Box Damper ☐

- **ID Fans**
  - Variable Frequency Drive ☒
  - Variable Frequency drive and inlet box damper ☐
  - Hydraulic Coupling/Variable Speed Drive ☐

Note: Scheme of fan control (e.g. Variable frequency drive etc.) to be decided specifically for each unit with the unit/ facility designer.

8.10.8 Burners

- Burners shall be Low NOx type.
- Continuous self-inspirating, fuel gas fired pilot shall be specified for each burner. Provision of portable igniters (one electric operated and one battery operated per cell) shall be kept.
- Metallurgy of pilot gas line from the FG header should be stainless steel.
- Minimum height of burner plenum from grade shall be 2100 mm
- Burners should be designed for entire range of the fuel specified in the datasheet.
- Flame scanner for pilot and main burner. Ionization rod to be provided for pilot burner and UV scanner for main burner.
- Noise Level: 85 dB(A) at 1m from source.
- Convection section tube type shall be finned tubes for gas firing.
- Burners are to be designed for max _____ mole% H_2 in the fuel gas resulting due to various scenarios (e.g. PSA failure). Design shall be based on various fuel gas composition as specified (Refer Basic Engineering Design Data, attachment-1 of this document for max. H_2 content in the fuel gas).
- Burner Management System (BMS) to be considered for all the heaters.
- Emission

a) MOEF notification G.S.R. 186(E) dated 18th Mar-08 Permissible emissions are

Parameter	Maximum emission Limit (mg/Nm3)
NOx - Oil firing	350
NOx - Gas firing	250
CO - Oil firing	150
CO – Gas firing	100
Particulate matter - Oil firing	50
Particulate matter - Gas firing	5

b) For pollution monitoring, sampling nozzles shall be provided on the stack as per central Pollution Control Board guidelines:

- 4 number of nozzles for stack diameter ≥ 2m (6" NB size with 1" inside lining)
- 2 number of nozzles for stack diameter < 2m (6" NB size with 1" inside lining)

- Burner Excess Air Levels: Burner shall be specified for operation at following excess air levels (in addition to excess air levels to be used for heater design), for fuel gas firing:
  - Natural Draft : 10%
  - Forced Draft : 5%

8.10.9 Convection Section

- Gas firing: Finned Tubes
- Convection bank for BFW preheating, steam generation, and steam superheating shall be designed for dry run conditions wherever the BFW feed pump doesn't have an emergency drive backup.

8.10.10 Standard Fired Heater Requirements

- Low-Low fuel gas pressure shuts down fuel gas supply but keeps pilots running.
- Low-low heater pass flow shuts down fuel gas but keeps pilots running.
- Emergency shutdown shuts down fuel gas as well as pilots.
- Low-Low fuel gas supply pressure shuts down fuel gas supply and return
- Low-low differential pressure between atomizing steam and fuel oil shuts down fuel oil supply and return
- Emergency coil steam introduction, manual or automated, depending on criticality
- Heater shuts down for the following:
    - Low-low fuel gas pressure shuts down fuel gas supply but keeps pilots running
    - Loss of Combustion Air
    - High-high Arch Pressure
    - High-high Outlet Temperature
- Draft gauge connections at: Burners, Below convection, Above stack damper, Below stack damper, at ID Fan suction
- Flue gas sampling connections at: Below convection section, Below stack damper
- On-line analysis with location as per connections mounted at the same plane: O2 analyser (TDLS type), NOx analyser, SOx analyser, CO analyser, SPM analyser.
- Minimum instrument should be provided as per latest OISD norms.
- Flue gas side Temperature measurement connections below convection section, below stack damper, at hearth level.
- Adequate number of flanged connections shall be provided for steaming out, water freeing, steam air decoking (where applicable) and chemical cleaning.
- Minimum metallurgy in hydrocarbon coils in heaters shall be as per process datasheet.
- Painting of stack shall be suitable for 8-10 yrs. of maintenance free operation.
- Skin thermocouples shall be considered for measuring temperature of furnace tubes (for each pass) as per OISD.
- Pad type skin thermocouples shall be considered for measuring temperature of furnace tubes. Two additional Type "B" thermocouples shall be installed close to the skin thermocouples for checking purpose. These Type "B" thermocouples shall be connected to an accessible terminal box for connection to a portable instrument.
- Damper should be operable pneumatically and manually from grade.
- Fuel gas lines shall always be traced (preferably electrical) from KOD up to individual furnace burners.

- In-line duplex strainers on fuel gas and pilot gas are recommended in burner piping for each unit. These shall be located not more than 20 meters upstream of the burner manifold and shall be 1 on-line + 1 spare strainer with mesh size of 200 for Fuel Gas and mesh size of 100 for Fuel oil. All mesh sizes are Tyler standard.
- Pilot gas line shall be provided with self-actuated PCV.
- Pilot gas line MOC: Stabilized Stainless steel (minimum).
- Main Fuel gas line MOC: CS (Carbon Steel) / Licensor specification as applicable.
- In case owner has any other preference, indicate here: –
  - Smart stack dampers (with dual sealing mechanism to avoid leakage) to be used as applicable.
  - Peep hole with glass to be provided in convection section as well (in addition to the radiant section)
  - Provision for addition of minimum 2 future tube rows (in the convection) to be provided
- Heater refractory requirements:

Type of Firing	Section	Refractory Type
Only Gas Fired	Radiant walls	Ceramic fibre
Oil or Only Gas Fired	Floor	Bricks
Oil or Only Gas Fired	Convection section	Castable

8.10.11 Coke Thickness for Hydraulic Calculations

Coke laydown shall take place in fired heaters handling hydrocarbons containing coke precursors. Heater tube diameter selection and hydraulics shall take cognizance of this in respective unit Licensor's Design Basis.

Return bends are preferred choice. Bends for the radiant section shall be inside firebox for radiant section and inside header box for convection section. Requirement of plug headers shall be specific to furnace service

8.11 LINE SIZING CRITERIA

Line sizing is performed for rated capacity. Unless there are specific Technology Licensor requirements, the line sizing shall be carried out as per criteria provided in the following sections.

8.11.1 Line Sizing Criteria for Liquids

SERVICE	LINE SIZE	MAXIMUM VELOCITY m/s	PRESSURE DROP (Kg/cm^2)/km	
			Normal	Max.
Pump suction, bubble point (1)	≤2”	0.6	0.6	0.9
	3” – 10”	0.9		
	12” – 18”	1.2		
	≥ 20”	1.5		
Pump suction, subcooled	≤2”	0.9	2.3	3.5
	3” – 6”	1.2		
	8” – 18”	1.5		
	≥ 20”	1.8		
Pump discharge, P ≤ 50 Kg/cm^2(g)		1.5 to 3 (8)	3.5	4.5
P > 50 Kg/cm^2(g)			7.0	9.0
Gravity flow		0.6 (3)	0.25	0.45
Side-stream draw-off (2)	≤ 2”	0.6	0.6	0.9
	≥ 3”	0.9		
Cooling water (7): i. sub-header		2.5	2.5	3.5
ii. main header			0.6 – 1.5	(4)
iii. Between pumping station and unit		1.5 to 3	1.5	(4)
Kerosene jet fuel		(5) 3.0 (max)		
Hot oil		1.0 (min)		
Lean Amine (velocity lamination max.)		2.0 for (CS) / 5.0 for SS		
Rich Amine (velocity lamination max.)		1.0 for (CS) / 5.0 for SS		
Caustic (velocity lamination max.)		1.5 for (CS)		

Notes:

(1) Applicable to liquid containing dissolved gas.
(2) Provide a vertical run of 3 meters minimum from nozzle, at nozzle size, before reducing the size of the line.
(3) Normal velocity.
(4) To be analyzed case by case.
(5) 50 to 100 m upstream tank inlet or loading facilities, the velocity shall be reduced to 1 m/s limit risks associated with static electricity.
(6) Pressure Drops for liquid to thermosiphon reboiler are 0.2 – 0.4 (Kg/cm^2)/km to be checked by heat exchanger section.
(7) Minimum velocity is 1 ~ 1.2 m/sec.
(8) For off sites large lines velocity criteria can be increased

8.11.2 Line Sizing Criteria for Gas & Vapors

	ρv^2 (max) Pa	Pressure Drop (Kg/cm^2)/km	Maximum Velocity m/s
1. VACUUM SERVICE		4 % Abs. Press. Max.	90
2. COMPR. SUCT.		0.2 – 0.7	
3. COMPR. DISCH. GAS		0.4 – 1.0	
4. STEAM (sub-headers)	15 000		(1)
a) 1 Kg/cm^2(g)		0.4 – 1.0	
b) 10 – 40 Kg/cm2(g)		1.0 – 2.0	
c) > 40 Kg/cm2(g)		(3)	
5. STEAM (long lines)	15 000		(1)
a) 1 Kg/cm^2(g)		0.1 – 0.2	
b) , c) > 10 Kg/cm2(g)		0.2 – 1.0	
6. KETTLE REBOILER OR NATURAL CIRCULATION RETURN LINE		0.2 – 0.4 (2)	
7. OVHD VAPOUR FROM STRIPPER		0.2 – 0.45 (2)	
8. COLUMN OVHD (P ≥ atm.)	15 000	0.3 – 0.6	
9. COLUMN OVHD (vacuum)		TOTAL = 5 mm Hg max	90
10. GAS LINES			(2)
P ≤ 20 Kg/cm2(g)	6 000		
20 < P ≤ Kg/cm2(g)	7 500		
50 < P ≤ 80 Kg/cm2(g)	10 000		
P > 80 Kg/cm2(g)	15 000		

Notes:

(1) The following table applies to set the velocity:

Pipe size	Saturated Steam	Superheated Steam
< 2”	10 m/s	15 m/s
3” – 8“	30 m/s	40 m/s
> 10”	40 m/s	60 m/s

(2) For large diameter lines of over 300 mm, it should be verified that the following velocities are not exceeded

Pressure (Kg/cm2(g))	Max. Velocity (m/s)
0.1 or lower	60.0
Up to 1.0	40.0
Up to 2.5	30.0
Up to 10.0	15.0
Up to 20.0	10.0
Up to 40.0	7.0

(3) To be analyzed case by case

8.11.3 Erosional Services

		Pressure Drop $(Kg/cm^2)/km$	Velocity m/s
1.	Mixed Phase Condensates	0.2 – 0.3	10 – 20
2.	Reboiler Return Line (Natural Circulation)	0.2 – 0.4	
3.	Partial Condenser Outlet	0.3 – 0.6	
4.	Mixed Phases at Compressor Delivery	0.4 – 1.0	

Vertical and horizontal pipes should not be affected by slug flow.

In a first step, the following criteria can be used

v_m = 10 to 23 m/s

$\rho_m v_m^2$ = 5 000 to 10 000 Pa (15 000 Pa Max)

where:

ρ_m = density of mixed phase

v_m = velocity of mixed phase (equal to 10 to 23 m/s)

8.11.4 High Velocity Services

In offshore facilities piping handling gaseous and mixed phase (gas / vapor + liquid) streams, velocity shall be lower than the erosional velocity defined in API RP-14E/3, that is:

$$V_e = \frac{C}{\sqrt{S}}$$

Where:

V_e = ft/s
S = lbs/ft³
C = refer to the following table

"C" VALUES	
FOR CONTINUOUS SERVICE	
150	For high alloy/corrosion resistant material.
100	For uniform corrosion with rates lower than 0.3 mm/a, or when a corrosion inhibitor is used.
80	For uniform corrosion with rates higher than 0.3 mm/a.
FOR DISCONTINUOUS SERVICE	
200	For all cases.

Velocity shall be in any case lower than 50 m/s in continuous services except vacuum services, while for discontinuous services (as safety valve discharges to flare) velocity shall be lower than:

- MACH for individual users 0.75
- MACH for flare header 0.50

In case of solid particle entrainment, limitations for discontinuous services shall be used.

8.11.5 Liquid & Gas with Solids

		Velocity (m/s)
1.	Liquid + solids	$1 \le v \le 3$ (1)
2.	Gas + solids (pneumatic conveying)	$20 \le v \le 30$ (1)

Notes:

1. For licensed plants, refer to Licensor requirements.

9. PIPING DESIGN

9.1 Inter Equipment Spacing

It is mandatory to adhere to stipulations of OISD standard 118 (latest edition) for minimum inter-equipment spacing and distance between process unit and process units & offsites.

For mounded vessels (mounded bullets), the minimum spacing of vessels and other safety regulations shall be as per OISD standard 150 (latest edition).

9.2 Piping

Engineering Design Basis shall define the detailed design requirements for overall piping, insulation, and painting and corrosion protection standards.

9.3 Heat Tracing Philosophy

Heat tracing philosophy for units (ISBL) and Offsite (OSBL) are listed below,

Heat Tracing Philosophy	**Yes/NO**
Within the unit battery limit, steam tracing shall be provided, (option of maximizing electrical tracing shall be considered).	Yes
In offsites, heat tracing shall be provided by:	
Steam tracing	No
Electrical tracing for temperatures up to 150 °C	Yes
Electrical tracing for temperatures up to 250 °C	Yes

LP steam shall be used for services with viscosity up to vacuum gas oil viscosity (pour point up to 70°C) and MP steam shall be used for heavy stocks.

LP steam tracing shall be used for service temperature up to 130°C.

Type of tracing shall be indicated in PIDs and Process Line List.

9.4 Guideline for Use of Gear Aided & Motor Operated Valves

- Gear aided operation shall be as per PMS.
- Gear aided valves shall be provided (of size 14” and above) for all the frequently operated valves, which can be opened or closed during normal unit operation such as isolation valves for pumps, compressors, filters etc., which are provided with standby or bypass.
- If Licensor requires motor-operated valves (MOV), the same shall be specified in P&ID (Process package)
- If Licensor requires gear aided valve (not specified in PMS), the same shall be specified in P&ID (Process package)

9.5 Special Item Identification

- All special purpose valves shall be tagged in P&IDs or identified in appropriate piping class. The Unit Design Basis shall contain appropriate specifications towards the same.

- All special piping items will be tagged and specifications will be furnished by Licensor / unit designer in the Process Package.
- Wherever two check valves (NRVs) are required to be installed in series to prevent reverse flow, the two check valves shall be of dissimilar type and from different manufacturers. This shall be duly noted on P&IDs.

9.6 Underground & Aboveground Lines

- All piping within a process unit shall be above ground except for oily water sewer, various closed blowdown systems and contaminated rain water sewer (CRWS).
- Cooling Water supply and return headers shall be on pipe rack for sizes up to 30". For sizes above 30" even though above ground routing is preferred, underground routing is also acceptable if there are plot constraints.
- All condensing vapor lines shall be specified with no pockets, free draining requirement.
- For flare line, isolation valve at unit B.L. should be lock open & it should be only in Horizontal line with stem pointing in downward (preferable) / horizontal position to avoid free fall of Gate & Blockage of line.
- For cooling water / raw water services, butterfly valves can be used including 12” and above. Gate valves to be used for lower sizes.
- All Wash Water/ Caustic / Neutralizing Amine / Filming Amine injection points in Column Overheads, Reactor Effluent lines and HHPS vapor lines shall have injection quills and spray nozzles.

10. INSTRUMENTATION

10.1 General Requirements

Following are the general requirements considered in basic engineering design:

- All symbols in P&ID shall be as per NRL Symbols and Legends.
- OISD 111 (Latest) and 152 (Latest) shall be complied with.
- Sensor / transmitter for shutdowns shall be separate from that of control / indication.
- Solenoid valve operation shall be considered for interlock shutdown system and on-off sequence valves like PSA / Dryers etc.
- Status indication shall be in DCS / auxiliary panel as applicable.
- Emergency Interlock bypass switches shall be hardwired only (Exceptions to be discussed with Owner / M-PMC)
- Hardwired recorders shall not be used.
- Shutdown signals shall be repeated in DCS from PLC.
- Instruments shall have individual tapings from process lines and equipment's.
- Critical rotating equipment's shall be provided with hard wired remote stop push button.
- Block and bypass valves to be provided for turbine meters, vortex PD meters, mass flow meters, magnetic flow meters and rotameters.
- 2 out of 3 voting required for all safety trip / ESD interlocks. In case of requirement for enhancement of any interlock not linked with safety shutdown, 2oo2 shall be considered on case to case basis.
- Rotating equipment with auto-start facility shall have auto-manual switch in Local Control Panel and at local control station (Field).
- No Process Switch (pressure, level, temp, and flow) shall be used.
- Instrument impulse lines to be insulated / traced, wherever required, to avoid condensation / congealing.
- Critical rotating equipment, pumps to have vibration monitoring system with control room Indication
- For PSA units, dedicated air filter to be provided for Instrument air header at inlet of the PSA skid.
- For Critical open loops, Closed loops, complex loops, final control elements etc., conventional non-field bus method (direct 4-20mA from / to field) shall be considered, and for non-critical open loops designated only for monitoring purpose wireless instrumentation system shall be considered, subject to approval from OWNER, for details refer to instrumentation design basis.
- For specific cases, online density meter to be considered for HC main feed to unit.
- Using of self-actuated pressure control valves shall be avoided except for tank blanketing and auxiliary system.

10.2 Flushing Requirement

It is recommended to use continuous flushing oil purge for flow, level and pressure transmitters in heavy residue, congealing services (except for bitumen product which can become off-grade) and services with fines for temperature above 200 °C in the vacuum service and for temperatures above 300 °C in the atmospheric service.

For instruments in vapor phase of congealing services e.g. in vacuum column instruments in vapor phase in the above temperature range, nitrogen purge / any other by Licensor / Designer shall be used.

10.3 Level Instruments

For Level Instrument type, nozzle size and connection details refer to Engineering design basis for Instrumentation, document number TP-1ZZZA-IC-BOD-0001.

10.4 Flow Instruments

For Flow Instrument type, nozzle size and connection details refer to Engineering design basis for Instrumentation, document number TP-1ZZZA-IC-BOD-0001.

10.5 Startup / Shutdown Operation from Control Room

Following emergency operations to be enabled from the control room:

- Individual unit emergency shutdown (control room Aux console and field)
- Fired heater emergency shutdown (control room Aux console and field)
- Critical rotating equipment shutdown (control room Aux console and field)
- All other rotating equipment shutdown (control room DCS and field)
- SOVs will have control room manual reset except heater related SOVs which will have field manual reset. However, licensor recommendation if any shall be incorporated.
- Shutdown valves to fail-safe position
- Damper (control room DCS and field)
- Gate operation (field only)

10.6 Control Philosophy for Vendor Package Items

All vendor package monitoring, control and logics shall be implemented in control room with minimum instrumentation in local control panel for startup and shutdown.

Governor Control Systems and Anti Surge Control Systems, which will be mounted on hardwired consoles in control rooms. (For details, refer to Instrumentation Design Basis).

10.7 Standard Instrumentation

10.7.1 Packed Towers

- Local differential pressure indication for each bed (DPT only).
- Local differential pressure indication for total section (DPT only).
- Control Room differential pressure indication for each bed.
- Control Room differential pressure indication for total section.

10.7.2 Heat Exchangers & Air coolers

- Local & DCS Temperature Instrument (TI) at process inlet and outlet of each service.
- Local & DCS Pressure Instrument (PI) at process inlet and outlet of each service.
- DCS Temperature Instrument (TI) at the outlet of each Air Cooler element.
- DCS Temperature Instrument (TI) at cooling water outlet of each exchanger.

11. GENERAL SAFETY ASPECTS

11.1 Emergency Isolation valves

Emergency Isolation valves (EIV) shall be provided for equipment and piping to isolate possible release of hazardous material. The provision of EIV's is linked to the risk associated to hazardous inventories, equipment and operation.

- EIV's shall be designed considering (subject to change based on licensor's requirement).
- Pneumatically operated by ESD system (through solenoid valve)
- Either in open or closed position in normal operation (Control valve shall not be used as EIV)
- Quick closing / opening
- Cannot be provided with a bypass
- Manual trip activated by the operator (in field at safe distance or from central control room)
- Local manual rest on the solenoid to avoid uncontrolled opening of the valve when the conditions are not safe
- Fail safe design in case of air/power supply failure.
- Valve position indication via open / close limit switches connected to DCS or position transmitter connected to ESD.
- Fire resistant valve
- Instrument air bottle to prevent spurious closing to be designed for 3 strokes to be considered on case by case basis.
- EIV on dangerous service shall be ball type valve to minimize leakage.
- Dual solenoid with independent signals from the ESD shall be provided for all shutdown valves in 2oo2 configuration in order to increase the reliability to prevent spurious trip.

11.2 Automatic Isolation valves Between Process Vessel & Pumps

Inventory isolation valves (solenoid operated shutdown valves (SDV)) will be provided as follows. All these valves and associated accessories to be fire safe type. Inventory isolation valves (Solenoid operated shutdown valves) will be provided for the following:

a) Pumped Circuits

- C4 and lighters pump suction irrespective of inventory
- C5 and heavier pump suction (50m3 and more inventory)
- Pumped liquid > auto-ignition irrespective of inventory
- Pumped liquid temp >250 deg C

b) Non-Pumped Circuits

- C4 and lighters irrespective of inventory (case to case basis)
- C5 and heavier irrespective of inventory (50 m3 and higher inventory on case to case basis)
- Fuel supply line to Heater
- For debutanizer / stabilizer bottoms being routed to OSBL directly or through other equipment, column bottom low level shutdown valve to be provided.
- HP to LP service to avoid high gas break through (These should be typically actuated by low low level in upstream Vessel).
- Rich Amine going to OSBL.
- Sour water going to OSBL

c) For the SOVs as described above, following to be considered:

- Start permissive to be provided for the pump with remote operated valve in open condition.
- Remote operated valve to be provided with limit switches indicating full open and full close condition in DCS. Pump to trip when remote operated valve is not in fully open condition.
- Remote operated valves to be provided with field mounted open and close switches. These switches to be installed at least 15 m away and at easily accessible location.
- Close switch to be also provided in DCS hardware console.

d) Process alarms shall be judiciously prioritize (emergency/high/low etc.) on the basis of criticality.

11.3 Constraint Due to Streams Containing H2S

When a process unit contains streams with H_2S content (Sour water) higher than or equal to 10 ppm wt., the following precautions shall be taken into account to avoid release of H_2S to atmosphere:

- Sample connections shall be with a closed loop.
- Closed sampling system shall be provided for sour water streams at outlet of HP Separators
- Draining of pressurized liquids which could release H2S after expansion shall be sent to a dedicated system as below
  - For hydrocarbons, Closed blowdown drum
  - Acid waters to sour water treatment system.
- Adequate H2S detection system in the process unit shall be provided.

11.4 Other Safety Aspects

Push buttons for stopping critical motors, shall be provided at a safe location, away from the fire-zone of the respective motors and also in control room. This is in addition to usual start stop push buttons provided for such motors. Critical motors to be identified based on Licensor requirement. Air cooler stop push button to be located at grade level in addition to start / stop button near the fan at platform.

Suction / discharge valves and suction filter should be close to the pump in order to avoid hydrocarbon wastage during filter cleaning and pump maintenance.

Flare line isolation valve at unit battery limits should be installed only in Horizontal line with stem pointing in downward (preferably) / horizontal position to avoid free fall of gate & blockage of flare line. Flare header leaving unit shall be provided with such isolation valves to facilitate maintenance of flare piping and pressure relief valves within the unit.

Emergency trips specified for HT and Critical motors shall be wired directly to switch gear in addition to control room.

Insulation shall be provided for personnel protection where operating temperature exceeds 60°C.

Furnace box / coil purge valve to be 15 m away from the furnace.

MB Lal Committee recommendation to be followed for Tank farm Area:

- In tank farm area, TSVs should be provided at operating manifolds outside the dykes.
- In tank farm area, hammer blinds for tanks isolation should not be considered.
- MOVs are to be provided on all outlet lines of Class A & Class B petroleum storage tanks, crude, finished and intermediate tanks. The first body valve on the tank is to be remote operated (MOV / ESD / Shutoff valve).
- MOV push buttons are to be located outside the dyke. It should be pull and push type.

- Provision of closing all tanks MOVs and ESDs are to be given from the control room. ESD's and MOV's are to be fire proofed
- Remote shutoff of MOVs from Control Room
- Tank Manual Dyke valves to be provided with position indicator (open / close) in Control Room.
- Tank handling A class product shall be designed with double shoe shield / Rim seal fire protection.
- Closed sampling system to be provide for Fluid class A, class B, Crude, Sulphur, H2, Toxic and Hazardous services
- HC Detectors to be considered as per MB Lal committee recommendation and OISD.
- Dual Level gauging system (one is Radar) with additional over fill protection shall be provided.
- Column handling hydrocarbon above auto ignition at its bottom will have isolation facility by way of remote operated valves at column bottom.

Double isolation valves: Double manual isolation valves shall be provided for equipment and instrumentation in following services which can be taken out of operation for inspection / maintenance with the unit still in operation. Typical examples such as spared pumps, control valve manifolds with bypass, spared relief valves and pressure indicators.

Double isolation shall be provided for Instruments in the following cases:

- For all service with rating >= 600#.

Double valving shall be provided for Equipment's / Pipe lines in the following cases:

- Hydrogen services with rating ≥ 600#.
- Rating ≥ 900# (even if not hazardous).
- Operating vents & drains on LPG service where flashing and consequent freezing can occur.
- Preventing product contamination where the use of spectacle blinds is not suitable.
- Dirty or coking services.
- Sour / Toxic services. (Note 1)
- Utility lines permanently connected to process lines or equipment in all cases.

Note 1: Double isolation valves requirement shall be followed as per Licensor recommendations. In case of absence of any such requirement/recommendations from Licensor then below guidelines shall be followed

Sour service:

Gas containing >24% H2S by Volume
(or)
Liquid containing > 4.0 psia H2S partial Pressure at operating temperature.

Toxic service:

Licensor recommendation to be followed.

12. SAFETY VALVES

12.1 Types of Safety Valve

Safety valves shall be balanced for discharge into closed system (flare header) and conventional for discharge to atmosphere. Pilot operated relief valves can be considered on a case by case basis in clean service (non-fouling & freeing) if process operating pressure is close to the relief valve set pressure.
Lines filled with liquid which may be blocked-in between two isolation devices shall be protected by safety valves for liquid expansion.

12.2 Safety Valve Sizing

Safety valves shall be sized according to API, ASME standards, OISD STD 106 and IBR requirements. For each safety valve, the applicable causes for relief and the resulting relief requirements shall be determined.

The following general emergencies shall be considered separately, unless one emergency will precipitate the other:

- Total cooling water failure
- Total electricity failure - Partial electricity failure, that is failure of one cable or transformer
- Steam failure, total or partial
- Instrument Air supply failure
- External fire, for each probable fire area

12.3 Setting of Single & Multiple Safety Valves

Refer to API RP-520, OISD STD 106 and IBR requirements

Pressure relief valve configuration is as follows:

- PSV for operational failures shall have installed spare with isolation facilities.
- PSV need not be equipped with a by-pass when depressurizing of the vessel to the flare is possible through accessible valves and suitable piping.
- PSV in hydrocarbon service will normally release to Flare.
- PSV exclusively for non-operational failures (external fire or heat exchanger tube rupture or thermal expansion) shall not have installed spare.
- If fire case is the governing case where process also requires safety valve, full capacity spare valve shall be provided.
- In general, PSV shall have upstream and downstream isolation valves.
- PSV in IBR steam service shall have no isolation.
- There shall be TWO full flow safety valves for steam generation equipment.
- PSV connected to open system shall have only upstream isolation (other than IBR)
- PSV on spare equipment shall not have spare.
- Dual safety valves on all pressurized storage vessels as per PESO requirement.
- PSV sizing and selection shall be as per guidelines provided in API 520 (parts 1 & 2), API 521, API 526, API 2000, ASME section-1 and IBR design requirements.

12.4 Basis of PSV Overpressure

Overpressure (as percentage of set pressure) for sizing relief valves shall be,

PSV Overpressure table:

S. No.	Contingency	
1	Steam generating / consuming equipment	3% or 5% as per code requirement
2	Fire case	21%
3	Thermal relief (Piping / Equipment)	25% / 10%
4	Operational failures	10%

13. CONTROL VALVES

13.1 Pressure Drop

Unless established by process requirements, control valve pressure drop values shall be calculated as the sum of the following two items, at normal flow rate:

- 20 % of circuit pressure drop, excluding the valve.
- 10 % of the static pressure of the system which the circuit discharges into, for pressures up to 15 Kg/cm2(g) (220 psi g); 1.5 Kg/cm2(g) (22 psi g) for pressures from 15 Kg/cm2(g) (220 psi g) to 30 Kg/cm2(g) (440 psi g); 5 % for pressures over 30 Kg/cm2(g) (440 psi g) (unless the pressure of the system which the circuit discharges into is directly connected with the suction circuit, e.g. reflux pumps). However, the following minimum values shall be specified at design flow rates: for

Control valves on liquid lines

- If upstream and downstream pressure are interdependent: 10 % of circuit pressure drop, or 0.7 Kg/cm2 (10 psi), whichever is greater.
- In all other cases: 5 % of pressure in discharge vessel, or 0.7 Kg/cm2 (10 psi).

Control valves on gas lines

- ΔP min = 0.2 Kg/cm2 (3 psi)

Control valves on reflux and recycle lines where static pressure variations influence the whole circuit shall be sized according to the same minimum values.

The DP for the closed valve shall be preliminary assumed as being equal to upstream design pressure, except for recycle lines such as reflux and recycle Hydrogen.

13.2 Flow-rate

As a general rule, control valves shall be specified for the following operating conditions:

- Maximum flow rate: 110 % of max. operating flow rate or according to connected equipment.
- Minimum flow rate: 40 % of minimum operating flow rate, in order to guarantee correct functioning at the extremes of the operating range.

Control valves shall also be checked in order that allowable noise levels are not exceeded.

13.3 Control Valve Materials

Control valve materials shall be selected according to line materials, taking into account valve operating conditions (flash).

13.4 Seat Leak Requirement

All the control valves discharging to atmosphere, flare, fuel gas or similar system must be specified as tight shut-off (TSO) type (ANSI Class V for liquids, VI for gases).

Control valves in services other than the above shall be specified with a seat leak standard.

On-Off valve connected to flare shall be TSO (leakage class V or better).

13.5 Control Valve Manifold

- As a default, control valves shall be provided with a manifold of upstream / downstream block and bypass valves. Type of bypass valve shall be same as main line control valve. Control valves not having block and bypass provision shall be provided with hand wheel for manual operation.
- The control valve inlet & outlet block valve size shall be same as corresponding line size. The bypass valve and line size shall be same as control valve size.
- Control valve sizes shall be as per Licensor / M-PMC engineering specification. Control valve selection to ensure that control valve shall be capable of operation between all specified flowrates from turndown to rated flow.
- Shut down valves shall not be provided with block and bypass valves.
- Anti-surge valve will not have bypass valves. Only ‘LO’ type isolation valve shall be provided.
- All the control valves shall have isolation valves, drain and bypass valve. Type of Bypass Valve shall be considered same as main line control valve.

13.6 Control Valve Bypass

Bypass shall be provided with the following exceptions:

- Control valve greater than 12” unless specified case by case.
- Normally no flow (NNF) control valve. (E.g. Start-up lines)
- No manual control is possible due to operational / safety reasons.
- Compressor anti surge valves (hand wheel not allowed).

For the above exceptions, isolation block valves, hand wheel (only if manual control is acceptable), drain valve shall be provided for the control valve manifold.

Size of bypass valve to be selected carefully if it has an impact on downstream relief valve (to be specified on datasheet for control valve and impacted relief valve).

13.7 Fail Safe Position

The choice shall be made so as to leave the plant in the safest possible condition following the loss of instrument air and / or electrical power supply:

Control valves which are usually designed to open on failure include (but not limited to):

- Reflux control valves
- Control valves in cooling duty
- Pump minimum flow control valve

Control valves which are usually designed to close on air failure include (but not limited to):

- Control valves on fluids supplying heat to the process
- Control valves on lines entering or leaving the unit
- Level control valves
- Pressure control valves

13.8 Control Valve Vent and Drain

All control valves shall be provided with ¾” drain valves both at upstream and downstream of control valve. For control valves in congealing hydrocarbon service, 1” bleeder shall be provided upstream and downstream of all control valves. This can be decided on case to case basis.

Drain valves shall be blind flanged for all hydrocarbon, hazardous and IBR steam service. Drain valves shall be capped for all non-hydrocarbon & non-IBR service. Control valves bleed / vent connected to flare shall be provided with double block valves.

14. RELIEF & FLARE SYSTEM

14.1 Relief System Design

Pressure relief systems shall be provided to ensure that equipment within the plant cannot be subjected to pressures beyond its design capacity. Flare headers and blowdown facilities shall be designed for all possible contingencies, discounting any process control or interlocks, which are designed to minimize any releases. The effect of auto-refrigeration during pressure relief shall be considered in the design.

All relief valves in flammable or toxic service shall discharge to the refinery closed flare system; discharge to atmosphere is not permitted other than under exceptional and controlled conditions.

Pressure relief valves shall be designed to accommodate 100% spare capacity to cater for mandatory testing requirements. Relief valves shall be interlocked to facilitate ease of removal and provide continuous on-line protection. Installation valves located between vessels and relief valves or located in relief headers shall be provided with CSO to ensure that they cannot be left in the closed position when the vessel or equipment they are protecting is live.

All relief valves discharging to a closed system and their discharge piping shall continuously slope towards the flare header.

14.2 Flare System

The flare system shall provide a safe collection and disposal route for all hydrocarbon relief and depressurizing flows from the refinery facilities during normal or upset conditions. The relief flows which are typically flammable, toxic and or corrosive, are converted to less objectionable compounds by combustion at elevated flares.

The Flare of NREP shall be connected to dedicated new flare. No interconnecting to the existing refinery flare is envisaged.

There will be dedicated flare systems provided for the following types of flare,

- Hydrocarbon Flare
- Acid Flare

Acid Gas Flare System shall be considered to manage the disposes of the releases containing acid gas from units which include Sulphur Recovery Unit (SRU), Amine Regeneration Unit (ARU) and Sour Water Stripper Unit (SWS).

14.3 Flare System Design

- Vapor releases of all molecular weights to be connected to flare system.

- For adherence to OISD 106, it is recommended that individual unit shall be provided with a flare KOD, whenever significant liquid relief is anticipated from the pressure relieving devices, apart from the main knockout drums at the flare stack. The flare KOD shall be sized to separate out liquid droplets down to a size of 400 microns and liquid relief continue for 15 min as minimum.

- OSBL main flare KOD shall be sized to separate out liquid droplets down to 600 microns.

- API 521 standard shall be adopted for estimating the governing flare loads.
- Flare knock-out drum provided at individual units to be horizontal.
- Purge gas (nitrogen) connection shall be provided in all dead ends of flare header in addition to the normal fuel gas connection for purging. Both Nitrogen and fuel gas connections shall be provided with rotameter for monitoring the flow.
- Flare header shall be sloped towards flare knock-out drum, slope value to be fixed as 1:500.
- Flare headers shall be provided with locked open double isolation block valves with bleed and spectacle blind at individual unit battery limits. These isolation valves should be installed only in Horizontal line with stem pointing in downward (preferably) / horizontal position to avoid free fall of gate & blockage of flare line.
- Continuous flow measurement systems to be provided in common stack (OSBL).
- Ultrasonic flow measurement will be provided for flare releases from each unit (ISBL).
- Main flare header commissioning / De-commissioning provision is to be kept.
- Flare water seal drum drain to be diverted to OWS.
- The flare header shall be designed for Mach number of 0.5.

Flare KOD to be equipped with automatic condensate pump out facility designed to handle lightest liquid hydrocarbon likely to be released.

15. STEAM & CONDENSATE SYSTEM

15.1 Introduction

This document describes the philosophy of the condensate collection and blowdown handling for all process units of the refinery. The system is provided to collect and return steam condensate for use as boiler feed water makeup while blowdown as cooling water makeup to minimize both water and energy consumption.

The designer / Licensor shall adopt this default philosophy towards sizing of equipment generating or consuming Steam:

- Steam turbine drives and ejectors shall be rated based on minimum steam conditions available at unit battery limits. (ISBL pressure and temperature drop to be considered).
- All other steam consuming equipment including heat exchangers shall be thermally rated based on normal steam conditions.
- All steam generating equipment shall be rated based on delivering steam at maximum conditions to the header.
- Normal steam conditions shall be used for operating utility estimates, heat and material balances and battery limit connections.

Condensate recovery will be maximized from process consumers. Segregated / separate system shall be considered for clean condensate and suspect condensate.

15.2 Condensate Control for Steam Consumers

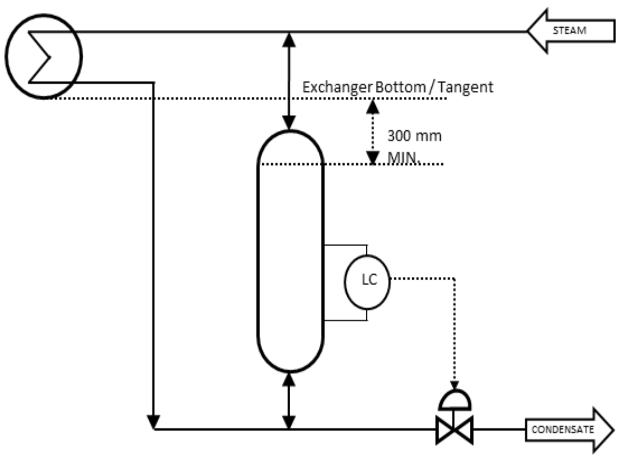
For steam flows up to 1000 kg/hr to a steam-heated exchanger, steam traps shall be considered for condensate removal. For larger consumers, condensate shall be collected in a pot provided with a level control valve for condensate withdrawal. The condensate pot top elevation shall be 300 mm lower than the exchanger bottom tangent line unless warranted otherwise from a process control scheme consideration.

The residence time for condensate in condensate pot shall be 90 seconds minimum.

The Steam condensate handling system is provided to collect and return steam condensate for use as boiler feed water make-up to minimize both water and energy consumption.

Steam condensate is returned in three forms as Suspect Condensate, Pure condensate (NOC), Cold condensate and Tracers/Trap condensate.

Typical Condensate Pot control scheme:



15.3 Suspect Condensate

Suspect condensate is condensate from exchangers and other equipment's where any leakage that might occur would be from the process into the steam / condensate (steam pressure lower than process pressure).

The equipment leak case is remote and unexpected. The steam condensate outlet shall be provided with hydrocarbon analyzer with DCS alarm (high) for leak detection. Suspect condensate is continuously monitored by online analyzer and normally routed to pure condensate collection system.

In case of hydrocarbon analyzer high alarm, Operator shall route condensate to suspect condensate collection system (refer Figure 1). The suspect condensate collected is pumped to condensate polishing (CP) unit for treatments.

15.4 Pure Condensate

Pure condensate is condensate from exchangers and other equipment where any leakage that might occur would be from the steam / condensate system into the process (steam pressure higher than process pressure).

Pure condensate is directly reused for steam generation.

HC analyzer and online conductivity analyzer with DCS alarms shall be provided in pure condensate return header from each unit within ISBL.

15.5 Cold Condensate

Cold condensate is condensate from turbines, which is clean condensate with respect to hydrocarbons, but can be contaminated with salts from cooling water and oxygen from ingressed air.

Surface condenser condensate will be routed normally for steam generation and to CPU (if Suspect with high conductivity) and shall be provided with online conductivity analyzer with DCS alarm (high).

15.6 Tracers / Trap Condensate

Tracers / Trap condensate is condensate from line steam tracing, process steam trap, line steam trap and storage tank / vessel heating coil. Tracers / Trap condensate shall be considered for condensate polishing if feasible. The non-recoverable steam condensates such as line steam trap and steam tracer trap shall be connected CRWS.

15.7 Steam Generator Blow Down Drains

Flash MP and HP blowdown for recovery of LP steam (if adequate steam generation is feasible) and further cool blowdown to 55 °C in cooling water exchanger and route to integrated Effluent Treatment plant for recycle.

15.8 Condensate Collection & Transfer System

The steam condensate recovery shall be maximized from process consumers. High / Medium pressure steam condensate shall be utilized for generating LP steam as much as possible.

HP and MP condensate form the steam consumers are flashed in a vessel. LP steam generated is routed to LP steam header and LP condensate liquid is further flashed to atmospheric pressure at condensate drum. The vapors are condensed in a vent condenser to 90 °C and returned to condensate drum. The total condensate from condensate drum is pumped to OSBL condensate header / utilized inside the process unit.

Pure and Suspect condensate routing shall be separate system (refer figure 1).

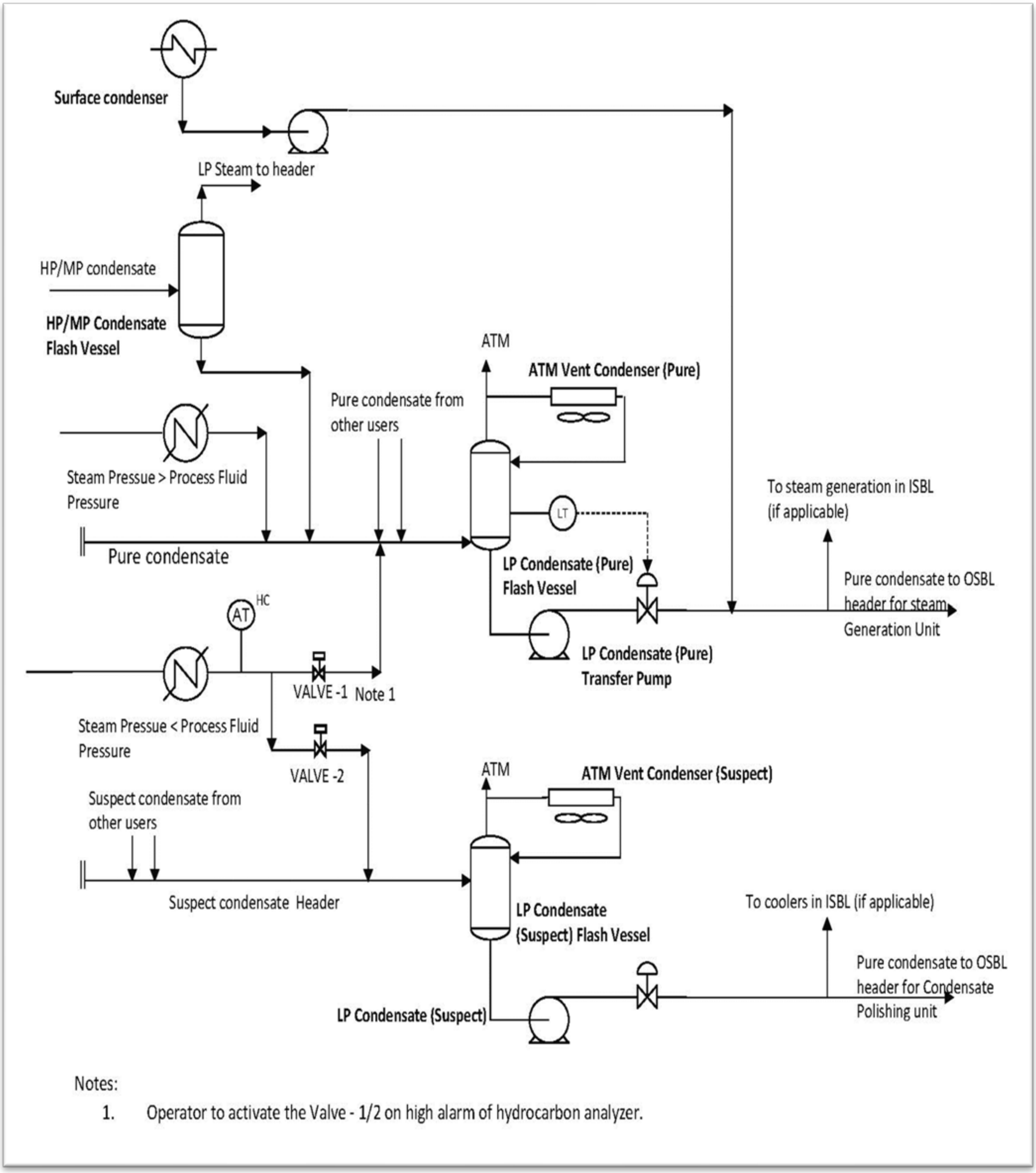


Figure -1 Typical Steam Condensate Collection System

16. IBR REQUIREMENTS

Scope of IBR

Steam generators / steam users shall meet the latest IBR regulations. Major IBR requirements are summarized below:

a) Vessels: Any closed vessel exceeding 25 liters in capacity which is used exclusively for generating steam under pressure and include any mounting or other fittings attached to such vessels, which is wholly or partly under pressure when steam is shut-off.

b) Piping: Any pipe through which steam passes and if:
   - Steam system mechanical design pressure exceeds 3.5 kg/cm2(g)

   or
   - Pipe size exceeds 254 mm internal diameter and having design pressure 1.0 kg/cm2g and above.

c) The following are not in IBR scope
   - Steam Tracing
   - Heating tubes in tanks
   - Steam Jackets
   - Heating coils

d) All steam users (heat exchangers, vessels, condensate pots etc.) where condensate is flashed to atmospheric pressure i.e. downstream is not connected to IBR system are not under IBR and IBR specification break is done at last isolation valve or equipment flange upstream of equipment.

e) All steam users where downstream piping is connected to IBR i.e. condensate is flashed to generate IBR steam are covered under IBR.

f) De-aerator, BFW pumps are not under IBR and IBR starts from BFW pump discharge.

g) Safety valves / Relief valves on IBR system will not have isolation valve.

` **Material Certificate:** (This section is only for information on IBR requirement).

- All items which are part of steam piping i.e. pipes, valves, fittings, traps, safety valves must have material certificates, countersigned by the local boiler inspectors.
- For imported items – Certificates issued by an authority empowered by Central Boilers Board (As per IBR) or under the law in force in a foreign country in respect of boilers manufactured in that country may be accepted.
- All drawings coming under preview of IBR shall be certified by Local Boiler Inspector.
- All datasheets of equipment which come under IBR scope shall have the annotation: "IBR is applicable".

17. DRAIN SYSTEM PHILOSOPHY

This document covers the design and operating philosophy for safe handling and disposal of liquid blowdown from various process equipment in the NREP. The closed blowdown and sewer system shall be designed as per OSID Standard 109, latest edition.

In this document," blowdown" shall refer to a liquid stream containing water, oil, chemicals or a combination of two or more of these which are required to be drained from various process equipment under different operating situations like start up, shutdown, normal operation or emergencies.

Continuous draining from process vessel to open funnel shall not be provided due to safety concern.

17.1 Hydrocarbon Drain

All drains from equipment / instruments handling hydrocarbon shall be connected to Closed blow down network(CBD), which shall be buried and lead to a closed blowdown drum, floating on flare pressure. The CBD discharge shall be pumped to slops tank in offsite area and shall not be put into sewers. Vapors from CBD vent shall be directed to flare system.

CBD drum shall be located in RCC pit and sand packed by Vibro compression with lean concrete on top 100 mm thick. The pit shall have a minimum clearance of 500 mm all around the vessel.

Adequate provision shall be provided prevent ingress of liquid into pit and remove any hydrocarbon / liquid accumulation in the pit. The CBD header and drum hall have suitable external protective coating to prevent it from corrosion. Cathodic protection is not required for the buried drum.

17.2 Sizing of Closed Blowdown Drum and Pump

- Dedicated Closed Blowdown drum shall be provided for individual units.
- Closed Blowdown drum shall be sized of minimum of 10 m3 to cater residual drains or to be sized for single largest equipment inventory.
- CBD pump shall be sized adequately to pump out the liquid in maximum of two hours. However, CBD pump of minimum capacity of 10 m3/h for units shall be provided.
- The liquids from the CBD drum is pumped out by pump vertical submersible pumps (1 working+1 ware house spare) which shall be automatic starting and stopping facility on high and low liquid levels in the drum.
- In case of units where unit flare KOD contents are routed to CBD, at least two pumps with independent power supply shall be provided.
- CBD system is designed for 200°C assuming only cooled liquid drains. Intermittent high temperature drainage can be tolerated by system.
- At the dead end of the CBD header steam and/ or inert gas connections shall be provided for proper purging and cleaning of the header. Drum shall be designed for steam out / vacuum condition.
- Underground drain line slope requirements shall be as per civil design basis and shall be designed to prevent any vapor reaching / flashing in the drum.
- Heavy hydrocarbons which tends to solidify under ambient conditions is classified as congealing (Black / Heavy) fluids and lighter hydrocarbons such as Naphtha, Kerosene and Diesel are classified as Non-Congealing (white / Light) fluids.
- Normally CBD drum shall be provided with cooling coils. If congealing hydrocarbon drains combined with non-congealing drains, combined cooling and heating coil to be provided in CBD drum.

- Separate CBD drums with pumps to be provided for collecting and transferring congealing and non-congealing hydrocarbon drains.
- LPG is drained directly to the flare system.
- Small quantities (check point) and base plate drain of pumps are connected to OWS.
- External corrosion allowance of 1.5 mm shall be considered for buried drain drums.

17.3 Amine Systems

- Amine drains from equipment handling amine solution shall be routed to a dedicated underground Amine blowdown drum (ABD) floating on acid flare and shall not be routed to any other hydrocarbon CBD system.
- Amine blow-down drum shall have oil separation & disposal facility (if required) / LP Steam sparer to remove dissolved H2S, if required.
- Amine blowdown drum shall be provided with submerged pump. Liquid from this drum shall be sent to ARU for reprocessing.
- For design requirement of ABD shall be similar to hydrocarbon CBD.
- Baseplate drains of pumps are connected to OWS.

17.4 Process Sour Waters

- Process sour waters from the units shall normally be routed to sour water stripper.
- Sour water within sour water stripper unit which will have large inventory of sour water will be provided with sour water blowdown drum (floating with acid gas flare system) for any drains.
- Provision for oil disengagement in the drum maybe provided if required and intermittently drained to CBD network.
- The residual low holdup drains (very infrequently) operation during maintenance only will be drained in OWS after dilution.
- Residual drains from instruments may be drained to oily water sewer with adequate safety precautions
- Small quantities and baseplate drain of pumps are connected to OWS. All baseplate drains of pumps are connected to the OWS sewer system.

17.5 Caustic Drain

All bulk caustic inventory drains shall leave process unit under own pressure or be pumped out after degassing. For residual caustic drains such as un pumpable vessel bottoms, level gauge drains, etc.

Drain Philosophy	Yes/No
Provide underground caustic CBD system with buried vessel and pump out	Yes*
Collect residual drains by temporary facility like drums and jars in case of small quantity	No
Collect residual drains by temporary facility and route to neutralization pit. From neutralization pit, it will be sent to OWS.	No
In process unit area, un pumpable caustic shall be recovered in an underground caustic collection vessel and then it will be sent to ETP	No
Drain residual caustic to OWS and flush with clean water	No

* - Common caustic drain system shall be considered on case to case basis.

17.6 Chemical Drain

Chemicals used in process / utility units shall not be drained to a sewer system and shall be collected in curbed / diked area and disposed safely. Disposal method shall be decided case by case.

Chemicals / RO reject water from DM water unit shall be routed to neutralization PIT and pumped to ETP / CRWS.

17.7 Steam Generator Blow Down Drains

Blowdown from a boiler / waste heat steam generator is done usually at high temperature and high pressure to maintain the water quality for steam generation.

MP or HP blowdowns from steam generators are flashed to low pressure for LP steam recovery and further cool blow down to 55°C in cooling water exchanger and route to CRWS.

17.8 Circulating Fresh Cooling Water Drain

- Drains from cooling water users shall be routed to CRWS.
- Fresh Cooling tower continuous blow down will be connected to ETP directly.
- Water from cooling water back flush shall be routed to CRWS.
- Discharge of TSV shall be routed to CRWS.

18. FLUSHING OIL SYSTEM & SLOP HANDLING

The slop and flushing oil system of NREP can be distinguished as:

18.1 Flushing Oil Systems

18.1.1 Normal Flushing Oil (FLO)

This is usually gas oil or procured diesel, charged to a header at a moderate pressure, for flushing out equipment, piping and instruments handling congealing liquids. FLO is stored in offsites, either separately or combined with product storage.

Centralized flushing oil system is envisaged

Flushing Oil System Design

- When FLO used as external flush for pump API seal plans, or for purging instruments in congealing service, the FLO needs to be boosted, the pressure becoming known only during detailed engineering.
- The flushing oil requirements are to be assessed, suitable scheme to be considered as applicable during design. During normal operation FLO header shall be pressurized through product run down lines of process units with strainer. Wherever continuous flushing such as pump / instant flushing is required, unit wise system shall be provided with individual vessel and pumps.
- Normally FLO booster system design will be considered (in individual unit) with surge vessel and pump where FLO received from offsite is stored in an adequately sized surge vessel and pumped to required higher pressure with in unit.
- Normal FLO pump and booster FLO pumps drives shall be motor driven.

18.1.2 Heavy Flushing Oil (HFLO)

This is usually straight-run vacuum gas oil cut required only in refinery units having hot, heavy fluid services. HFLO is not stored in offsites because only one cut of vacuum gas oil is selected whereas offsite storage is a mixture of multiple cuts. A centralized system is usually located in the originating or largest consuming unit (to be finalized), charging to a header at a pressure to be determined during detail engineering and at a temperature of around 80 - 110 °C. This HFLO can be used as external flush for those pump API seal plans where FLO cannot be used because of vapor blowout possibilities inside the seals.

During normal operation, HFLO shall be sourced internally within process unit. During startup / shutdown, hookup connection from appropriate external source shall be decided during detail engineering.

18.2 Slop Handling

18.2.1 Wet Slops

Slops originating from CPI, OSBL drain vessels that can contain large quantities of water. Wet slops are routed to the Light slop tank blanketed with nitrogen.

18.2.2 Light Slops

Flushing oil return from ISBL units handling light hydrocarbons (Diesel and lighter) is routed to the light slops header. The slops from these units, including the slops from the light HC drain vessels are routed to Light slops tank. The water phase is drained under gravity to the OWS, while the oily phase is pumped under flow control to the CDU unit (or to the suction of Crude charge pump) The dry slops tank are also used during startup of the CDU / VDU to store (on spec / off spec) products from this unit.

18.2.3 Heavy Slop

Heavy Flushing oil return from ISBL units handling Heavy / congealing hydrocarbons (heavier than Diesel) is routed to the Heavy slops header. The slops from these units, including the slops from the Heavy HC drain vessels are reprocessed within these units. Any quantity which is not reprocessed within the respective units are routed to Heavy slops tank. The heavy slops tanks are equipped with an LP steam coil for heating the contents of the tank to prevent the segregation of light and heavy hydrocarbons and keep the hydrocarbon in flowing conditions.

19. ENVIRONMENTAL CONTROL

19.1 CPCB Emission Standards For Furnace & Boilers

a. Emission norms for Furnace, boilers and heaters (As per Air (Prevention and Control of Pollution) Act, 1981 and its subsequent amendments)

Parameter	Firing Mode	Limiting concentration (mg/Nm3)
Sulphur Dioxide (SO2)	Gas	50
Oxides of Nitrogen (NOx)	Gas	250
Particulate Matter (PM)	Gas	5
Carbon Monoxide (CO)	Gas	100

1) All the furnaces / boilers with heat input of 10 million kilo calories/hour or more shall have continuous systems for monitoring of SO_2 and NOx. Manual monitoring for all the emission parameters in such furnaces or boilers shall be carried out once in two months.
2) All the emission parameters in furnaces / boilers having input less than 10 million kilo calories / hour will be monitored in three months.
3) In case of continuous monitoring, one hourly average concentration values shall be complied with 98% of the time in a month. Any concentration value obtained through manual monitoring, if exceeds the limiting concentration value shall be considered as non-compliance.

b. Emission norms for FCC Regenerators (As per Air (Prevention and Control of Pollution) Act, 1981 and its subsequent amendments):

Parameter	Limiting Concentration (mg/Nm3) for New refinery units
Sulphur Dioxide (SO2)	500 (for hydro-processed feed) 850 (for other feed)
Oxides of Nitrogen (NOx)	350
Particulate Matter (PM)	50
Carbon Monoxide (CO)	300
Nickel and Vanadium (Ni+V)	2
Opacity, %	30

Notes:

1) In case part feed is hydro-processed, the emission values shall be calculated proportional to the feed rates of untreated and treated feeds.
2) FCC regenerators shall have continuous system for monitoring of SO_2 an NOx. One hourly average concentration values shall be complied with 98% of the time in a month, in case of continuous monitoring. Manual monitoring for all the emission parameters shall be carried out once in two months.
3) Any concentration value obtained through manual monitoring, if exceeds the limiting concentration value, shall be considered as non-compliance.
4) Data on Sulphur (wt%), Nickel (PPM) and Vanadium (PPPM) content in the feed to FCC shall

be reported regularly.

5) Limit of Carbon Monoxide emissions shall be complied within except during annual shutdown of CO boiler for statutory maintenance.

c. Emission norms for Sulphur Recovery Units (As per Air (Prevention and Control of Pollution) Act, 1981 and its subsequent amendments):

Parameter	Capacity (tonnes/day)	Limiting values for New SRU
Sulphur recovery, %	Above 20	99.5
H2S, mg/Nm3		10
Sulphur recovery, %	5-20	98
Sulphur recovery, %	1-5	96
Oxides of Nitrogen (NOx), mg/Nm3	All capacity	96
Carbon Monoxide, mg/Nm3	All capacity	100

Notes :

1) Sulphur recovery units having capacity above 20 tonnes per day shall have continuous systems for monitoring of SO2. Manual monitoring for all the emission parameters shall be carried out once in a month.

2) Data on Sulphur Dioxide emissions (mg/Nm3) shall be reported regularly.

3) Sulphur recovery efficiency shall be calculated on monthly basis, using quantity of Sulphur in the feed to SRU and quantity of Sulphur recovered.

19.2 Aromatics Handling

Special precautions will be applied towards handling process streams containing aromatics. Threshold values for applying special precautions are:

- Benzene content greater than 1% weight.
- Aromatics greater than 25% weight.

The design and equipment provisions toward special precautions shall be:

- Sampling points as defined in standard legend P &ID
- Dual mechanical seals for pumps
- Connecting the following to the unit closed blowdown system as possible:

Requirements	
Vessel drains	Yes
Pump drains	Yes
Level gage, level instrument & Standpipe drains	Yes
Control valve drains	Yes
Drains of equipment (isolated for maintenance)	Yes
Special Gas detectors (e.g Benzene detectors) installed at strategic locations	Yes

19.3 General Considerations

- Low NOx burners shall be used in all furnaces.
- Emission from the stacks shall be as per the norms as stipulated by CPCB and other mandatory regulations.
- Emission from the project will be such that the National Ambient Air Quality standards are met outside battery limits.
- The hazardous waste generated shall comply with Hazardous Waste (Management & handling) Rules, 2016.
- Liquid effluent generated shall conform to MINAS standards.
- All unit designers have to necessarily furnish complete effluent characteristics as defined in the section 18.5
- **Noise level:**
  - 85 dB(A) at 1 meter from source
  - 75 dB(A) daytime and 70 dB(A) night time at the fence.
- **Relief /Depressurization:**
  - For PSV relief / depressurization allowable noise to be 115 dB(A)
  - For Control valve releasing to atmosphere, noise limit shall be 85 dB(A)
- Online Continuous Emission Monitoring System (OCEMS) shall be provided with Remote calibration facility. This is required for data verification / validation by SPCB’s / PCC’s.

19.4 Stacks

- Stacks shall be individually mounted on each heater unless there are considerations such as grade mounted APH or combined APH for group of heaters.
- Minimum fired heater stack height for various heater shall be higher of indicated heights in respective unit documents or as calculated in the formula below :
  - Height (H) of stack from Grade : 60 m or
  - Stack height using formula H=14(Q)0.3 (where H is stack height, meters and Q is total SO_2 emission, kg/hr) shall also be calculated and if it is higher than that indicated above, same shall be adopted.
- Top of stack shall be at least
  - 5 m higher than highest working platform within radius of 30 m
  - 3 m higher than highest working platform for radius of 30-50 m
- The maximum height of stack is not to exceed ___m (Not Applicable)
- For pollution monitoring, 6” NB size (with 1” inside lining) sampling nozzle shall be provided on the stack as per Central Pollution Control Board guidelines:
  - 4 nozzles for stack diameter >2.0m
  - 2 nozzles for stack diameter <= 2.0 m
  - 1 number isokinetic nozzle at 500 mm above and 90 °C to SPM analyser at stack.
- All sampling point should be accessible and adequate platform to be provided for the same.

19.5 Effluent Data

19.5.1 Gaseous Effluent Data

To be filled by Licensor / Detailed engineering Contractor

Description	Unit of Measurement
Source/Originating Point	
Type of Emission (Fugitive/Stack)	
Flow rate	m^3/hr
Type of Flow (Cont./Intermit)	
Frequency and Duration If Intermittent	
Height of Stack/Source	m
Diameter of Stack (TIP)	m
Temperature of Gases	°C
Pressure	kg/cm^2a
Mol. Weight of Gases	
Density of Gases	Kg/m^3
Viscosity of gases	cP
Exit Velocity of Gases	m/sec
Composition of Gases (WT%)	
Hydrocarbons (HC)	
CO (Carbon monoxide)	
CO2 (Carbon Dioxide)	
SO2 (Sulphur Dioxide)	
N2S (Nitrogen sulphide)	
SO3 (Sulphur oxide)	
NOx (Oxides of Nitrogen)	
N2 (Nitrogen)	
O2 (Oxygen)	
H2O (Water vapours)	
Any other component (Chemical Composition)	
Suspended particulates matter (SPM)	
Density & Bulk Density	Kg/m^3
Dust Load	g/m^3
Nature of Dust	
Amorphous	
Crystalline	
Needle shaped	
Fibrous	
Metallic	
Particle Size Distribution (wt%)	
Up to 5 microns	
5-20 microns	
20-40 microns	
40-80 microns	
Above 80 microns	

19.5.2 Aqueous Effluent Data

To be filled by Licensor / Detailed engineering Contractor

Description	Unit of Measurement
Source / Originating Point	
Flow Rate (Average/Peak)	m^3/hr
Frequency of flow	Continuous / Intermittent
Duration if Intermittent	Hours / day
Temperature	°C
Color/ Odor	
Pressure	Kg/cm^2g
Sp. Gravity	
Viscosity	cP
Vapor Pressure	Kg/cm^2a
pH	
BOD, 20 °C	mg/lit
COD	mg/lit
Suspended Solids	mg/lit
Sp. Gravity of solids	
Particle size	Microns
Total Dissolved solid	mg/lit
Volatile matter	mg/lit
Total Oil content	mg/lit
Viscosity of oil	cP
Phosphates	mg/lit
Sulfates	mg/lit
Chlorides	mg/lit
Fluorides	mg/lit
Nitrates	mg/lit
Total Nitrogen	mg/lit
Free/Fixed Ammonia	mg/lit
Cyanides	mg/lit
Sodiumates	mg/lit
Calcium	mg/lit
Magnesium	mg/lit
Phenolic Compounds	mg/lit
Heavy Metals	mg/lit
Chromium	mg/lit
Nickel	mg/lit
Lead	mg/lit
Zinc	mg/lit
Mercury	mg/lit
Any other	mg/lit
Emulsified oil	
Chemical Composition	% wt.

12.1.1 Solid Effluent Data

To be filled by Licensor / Detailed engineering Contractor

Description	Unit of Measurement
Source/Originating Point	
Quantity of Solid Waste	
Quantity of Solid Waste	Kg/day
Continuous/Periodical	
Frequency & Duration, if periodical	
Temperature	
Density	
Physical Condition (Slurry / Sludge/Powder/ Lumps)	
Viscosity, If slurry/sludge	
Nature of Waste (Hazardous/Toxic/Explosive/ Corrosive/Abrasive/Radioactive)	
Calorific Value	Kcal/kg
Chemical Composition	Wt%
Volatile Matter	Wt%
Moisture Content	Wt%
Recycling of Waste Required	Yes/No
Recovery of Valuable component	Yes/No
Schemes of Reuse/Recycle	
Sale value of Waste	Rs./Ton
Availability of Land for Disposal	Yes/No
Leachate Characteristics (If Slurry)	

ATTACHMENT-1

BASIC ENGINEERING DESIGN DATA

TABLE OF CONTENTS

1. LOCATION

Country	India
State	Assam
District	Golaghat
Town/City	Numaligarh
Plant site co-ordinates	Longitude 93º 43' 30" E & Latitude 26º 37' 30" N
Site altitude	90.4m above MSL

2. SITE DATA

2.1 Climatic Conditions

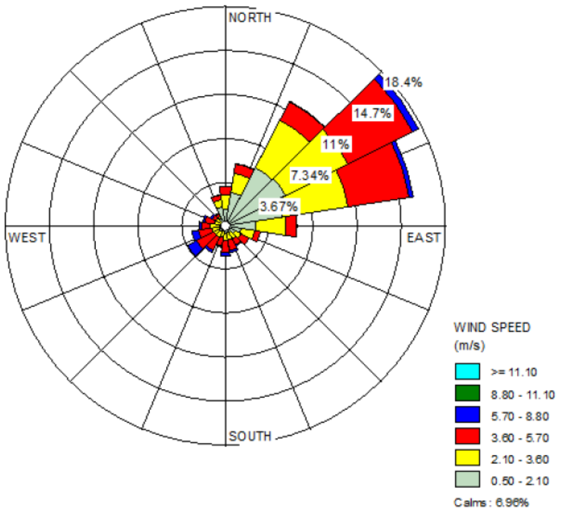
Ambient temperature: Normal Minimum Maximum Design	 35 °C 5 °C 38 °C 65 / 5 °C	
Design Minimum Temperature (MDMT):	5 °C	(Note 1)
Winterizing temperature:	5 °C	(Note 2)
Relative humidity:	85% at 35 °C	
Design dry bulb temperature: (for other design activities including HVAC)	35 °C	
Design wet bulb temperature: (for other design activities including HVAC)	27 °C	
Design Air temperature for air cooled exchangers where followed by water cooling	35 °C	
Design Air temperature for air cooled exchangers where not followed by water cooling	35 °C	
Design Air temperature for GT's / Boiler	35 °C	
Coincident relative humidity , % and temperature °C for Air Blower / Air Compressor/ GT / Boiler	85% at 35 °C	
Barometric pressure: Normal:	994.7 mbar(a)	
The design temperature for a surface exposed to direct sunshine	65°C	
Rainfall data for 1-hour period	90 mm (max)	
Rainfall data for 24-hour period	160 mm (max)	

	(rainy season is from April to September)	
Maximum Recorded flood level.	To be defined later	
Earthquake Design factor	(Code IS-1893 Zone-V/ Site spectra whichever is governing) As per IS 1893, Seismic factor Z = 0.36	

Note 1: Used as MDMT (Minimum Design Metal Temperature), unless otherwise specified.
Note 2: Used for tracing requirement.

2.2 Wind

- Direction of prevailing wind predominantly: As per wind rose, refer Civil Engineering Design Basis
- Velocity to be used for insulation: 36 km/hr (10 m/s) or as per IS-875 whichever is higher.
- Velocity to be used for Flare radiation calculation: 36 km/hr (10 m/s).
- Wind Velocity, max: 140 km/hr
- Wind Direction: NE to SW



Primary wind rose diagram

2.3 Environment Condition

- Corrosive Industrial Environment : Yes
- Extreme Moisture (Tropical Climate) : Yes
- Marine Exposure (salt spray) : No
- Exposure to pollutants originating from surrounding industrial plant : No
- Thunderstorm Frequency : Yes (High)
- Basic wind speed, Vb : 50 m/sec as per IS875 Part 3 : 2015
- Terrain category for K2 : Category 2, As per Code IS875 Part 3 : 2015
- K3 factor, Ka,Kc,Kd : As per Code IS875 Part 3 : 2015

3. SYSTEMS (UNITS) OF MEASUREMENT

This following table indicates the units of measurement to be used for the Numaligarh Refinery Expansion Project.

Parameter	Preferred unit	Alternate unit
Ampere hours	ampere-hours	
Angle	deg	
Apparent power	VA	
Area	m^2	
Composition -Gas/Vapor	Mol % or Vol%	ppmv
Composition -Liquid	Wt%	ppmw
Current Density	A/m^2	
Date	DD-MM-YYYY	
Electrical Conductivity	µS/cm	
Electrical Current	A	
Electrical Conductance	S	
Electrical potential	V	
Electrical resistance	ohm	
Enthalpy	Kcal/kg	
Entropy	kcal/kg-°C	
Flow Rate (gas)	Nm^3/hr	
Flow Rate (liquid)	m^3/hr, Sm^3/hr	Kg/hr (or) l/hr
Flow Rate (mass)	Kg/hr	T/hr
Flow Rate (steam)	Kg/hr	T/hr
Force	N	
Fouling resistance	m^2 °C hr/Kcal	
Furnace draft	mmWC	
Heat	Kcal	
Heat rate	MMKcal/hr	Gcal/hr
Heat transfer coefficient	Kcal/hr-m^2-°C	
Inertia	Kg/m2	
Length	mm	M
Liquid density	Kg/m^3	
Liquid relative density	Sp. Gr. T°C/15°C	
Mass	Kg	
Mass per length	kg/m	
Pipe diameter	in	
Plant Capacity	T/hr	m^3/hr, TM/MMTPA
Power	kW	MW
Pressure (absolute)	Kg/cm^2 (a)	
Pressure (gauge)	Kg/cm^2 (g)	
Pressure (vacuum)	mmHg	mmWC
Reactive power	VAR	

Resistance per length	Ohm/m	
Sound Pressure Level	dBA	
Specific heat	Kcal/kg °C	
Standard liquid	Sm^3/hr @ 15°C& 1.033 kg/cm^2 (a)	
Standard vapor	Nm^3/hr @ 0°C & 1.033 kg/cm^2 (a)	
Stress	N/m^2	
Storage tank pressure	mmWC (g)	kg/cm^2 (g)
Surface Tension	N/m	dynes/cm
Temperature	°C	
Thermal conductivity	Kcal/hr-m-°C	
Thermal Expansion	$°C^{-1}$	
Thickness	mm	
Time	hr	Sec, min
Torque	N-m	
Vacuum	mmWC	mmHg
Vapor Flowing density	Kg/m^3	
Velocity, linear	m/s	km/hr
Velocity, angular	rpm	
Vessel nozzle sizes	in	
Viscosity, Dynamic	cP	
Viscosity, kinematic	cSt	
Volume (gases/liquid)	m^3	

4. UTILITY CONDITIONS & SPECIFICATIONS

4.1 Utility Conditions

Unless and otherwise specified, utility conditions are at unit battery limits [All B/L pressure are as measured at grade]

S.No.	Parameter	Minimum	Normal	Maximum	Mech Design
1	**HIGH PRESSURE (HP) STEAM**				
	Pressure, kg/cm^2(g)	39	40	42	47 / FV
	Temperature, °C	415	454	460	480
2	**MEDIUM PRESSURE (MP) STEAM**				
	Pressure, kg/cm^2(g)	13.5	14.5	15.0	16.5 / FV
	Temperature, °C	215	220	225	240
3	**LOW PRESSURE (LP) STEAM**				
	Pressure, kg/cm^2(g)	3.5	4.0	4.5	6.5 / FV
	Temperature, °C	150	165	170	200
4	**CONDENSATE RETURN (PURE & SUSPECT / SURFACE)**				
	Pressure, kg/cm^2(g)	-	6.0	-	12 / FV
	Temperature, °C	-	90	-	150
5	**SERVICE WATER**				
	Pressure, kg/cm^2(g)	4.0	4.5	5.0	12
	Temperature, °C	Amb.	Amb.	Amb.	65
6	**DRINKING WATER**				
	Pressure, kg/cm^2(g)	4.0	4.5	5.0	8
	Temperature, °C	Amb.	Amb.	Amb.	65
7	**COOLING WATER (Note A)**				
	Supply Pressure, kg/cm^2(g)	4.5	5.0	5.5	10.5
	Return Pressure, kg/cm^2(g)	2.5	3.0	3.5	10.5
	Supply Temperature, °C	-	33	35	65
	Return Temperature, °C	-	45	45	65

8	**DM WATER**				
	Pressure, kg/cm^2(g)	5.0	5.5	6.0	12
	Temperature, °C	Amb.	Amb.	Amb.	65
9	**BOILER FEED WATER (HP / MP)**				
	Pressure, kg/cm^2(g)	-	60/ 30	65 / 31.5	85 / 44
	Temperature, °C	-	105 / 105	110 / 110	150 / 130
10	**FIRE WATER**				
	Pressure, kg/cm^2(g)	7.0	10.5	-	15
	Temperature, °C	Amb.	Amb.	Amb.	65
11	**PLANT AIR**				
	Pressure, kg/cm^2(g)	4.5	5.5	6.5	10.5
	Temperature, °C	Amb	Amb	45	65
12	**INSTRUMENT AIR**				
	Pressure, kg/cm^2(g)	4.0	4.5	5.5	10.5
	Temperature, °C	Amb	Amb	45	65
13	**NITROGEN**				
	Pressure, kg/cm^2(g)	4.0	4.5	5.0	9
	Temperature, °C	-	Amb.	-	65
14	**HYDROGEN (EXPORT FROM HGU)**				
	Pressure, kg/cm^2(g)	18.5	19.0	20.5	22.5
	Temperature, °C	-	45	-	150
15	**IGGL NATURAL GAS (Note B)**				
	Pressure, kg/cm^2(g)	35	45	55	92
	Temperature, °C	0	Amb.	Amb.	65
16	**REFINERY FUEL GAS (Note B)**				
	Pressure, kg/cm^2(g)	3.0	3.5	4.5	7
	Temperature, °C	Amb.	Amb.	Amb.	65
17	**SOUR WATER @ SWS B.L**				
	Pressure, kg/cm^2(g)	5.0	5.5	9.0	16.2
	Temperature, °C	-	51 - 55	-	85

18	**STRIPPED SOUR WATER**				
	Pressure, kg/cm²(g)	-	6.0	-	15
	Temperature, °C	-	40 - 65	-	100
19	**DILUTED CAUSTIC (15wt%)**				
	Pressure, kg/cm²(g)	-	4.0	-	8
	Temperature, °C	-	40	-	65
20	**SPENT CAUSTIC**				
	Pressure, kg/cm²(g)	-	6.0	-	13
	Temperature, °C	-	38	-	65
21	**RICH AMINE @ ARU B.L**				
	Pressure, kg/cm²(g)	-	4.5	-	16
	Temperature, °C	-	50	-	120
22	**LEAN AMINE**				
	Pressure, kg/cm²(g)	-	10.5	-	16
	Temperature, °C	-	45	-	120
23	**FLUSHING OIL**				
	Pressure, kg/cm²(g)	3.0	-	-	10
	Temperature, °C	40	-	-	65
24	**CLEAN CONDENSATE SUPPLY**				
	Pressure, kg/cm²(g)	-	5.0	-	12
	Temperature, °C	-	90	-	150

Note

A. CW Supply and Return temperature shall be used consistently. For design purpose the cooling water supply / return temperature shall be considered as 35 / 45 ˚C respectively.

B. **IGGL Natural gas is used for producing Hydrogen in HGU.** Refinery Fuel gas is consumed in all the furnaces within the process units, steam and power generation systems. Additional requirement is met by IGGL Natural gas.

4.2 Utility Specification

4.2.1 Service Water Specification

Quality	Units	Normal
pH		7.24
Total hardness as $CaCO_3$	mg/l	146.38
Calcium as Ca	mg/l	21.5
Magnesium as Mg	mg/l	22.54
TDS	mg/l	223
Total Alkalinity	mg/l	123
Total Suspended solids	mg/l	5
Chloride as Cl	mg/l	28.6
Sulphate as SO4	mg/l	12.8
Turbidity in NTU		
Conductivity	µS/cm	346
Iron as Fe	mg/l	0.13
Silica	mg/l	13-18

4.2.2 DM Water Specification

Quality	Units	Design
pH		8.5-9.5
Total hardness as $CaCO_3$	mg/l	Nil
Total Silica as SiO_2	mg/l	< 0.02 (max)
Total iron as Fe	mg/l	0.01 (max)
Total Copper, as Cu		-
Oil Content		Nil
Conductivity at 25 °C	µS/cm	< 0.2 (max)

4.2.3 Boiler Feed Water Specification

Quality	Units	Design
pH		8.5-9.5
Total hardness as $CaCO_3$	mg/l	Nil
Total Silica as SiO_2	mg/l	< 0.02 (max)
Total iron as Fe	mg/l	0.01 (max)
Total Copper, as Cu		-
Oil Content		Nil
Conductivity at 25 °C	µS/cm	< 0.2 (max)
Oxygen	mg/l	< 0.005 (max)

4.2.4 Cooling Water Specification

Quality	Units	Normal
pH		7-8
Total hardness as $CaCO_3$	mg/l	732
Calcium as Ca	mg/l	108
Magnesium as Mg	mg/l	113
TDS	mg/l	1115
Total Suspended solids	mg/l	50
Chloride as Cl	mg/l	143
Sulphate as SO4	mg/l	45
Turbidity in NTU		
Conductivity	µS/cm	1730
Iron as Fe	mg/l	0.50
Silica	mg/l	65-90

4.2.5 Nitrogen Specification

Quality	Design
Nitrogen Purity	99.99 vol% minimum
O_2 Content	5 ppmv maximum
CO_2 Content	1 ppmv maximum
CO Content	Nil
Quality	Dry & Oil free
Dew point	-100 °C
Oil Content	Nil

4.2.6 Plant Air Specification

Quality	Design
Dew Point (Ambient)	Saturated
Oil Content	Nil
Quality	Dust free

4.2.7 Instrument Air Specification

Quality	Design
Dew Point (Ambient)	-40 °C @ atmospheric pressure
Oil Content	Nil
Quality	Dust free

Note:

Instrument Air Buffer volume: Instrument air receiver holdup time shall be sized adequately for minimum 30 mins of supply after power failure to facilitate safe shut-down of the unit.

4.2.8 Refinery Fuel Gas Specification

Composition	Case -1 (mole %)	Case -2 (mole %)
H_2O	1.23	1.38
O_2	0.37	0.55
N_2	6.93	5.46
CO	NIL	NIL
CO_2	0.92	0.55
Hydrogen	23.37	27.84
Methane	32.42	34.71
Ethane	14.20	16.44
Ethylene	14.75	6.57
Propane	2.48	3.20
Propene	0.75	0.28
i-Butane	0.19	0.25
n-Butane	0.83	1.05
1-Butene	0.25	0.11
i-Pentane	0.00	0.00
n-Pentane	1.30	1.61
Total	**100**	**100**
H_2S ppm	**100 (max)**	**100 (max)**
LHV (Kcal/Kg)	10308	10636
Average MW	19.85	18.62

4.2.9 IGGL Natural Gas (R-LNG) Specification

Components	mole%
Water	Note-1
Nitrogen	0.028
Methane	93.670
Ethane	6.099
Propane	0.164
i-Propane	
n-Butane	0.019
i-Butane	0.021
H_2S, ppm, wt	5 (max)
Total Sulphur, ppm, wt	10 (max)
Specification	
LHV (Kcal/kg)	11860
Gas MW (kg/kmol)	16.96 (Note-1)

Notes:

1. Moisture content in natural gas is Nil. However, for design, pipeline spec of 80 kg moisture/ MMSCM of Natural gas to be considered.

From other source:

R-LNG from Dahej / Dhamra is proposed to be transported through JHBDPL/BGPL for gas injection point to the North-East gas Grid at Guwahati

Refer below table for Gas composition

Gas Components	mole%
C1	98.34
C2	1.5
C3	0.02
i-C4	0
n-C4	0
i-C5	0
n-C5	0
C6+	0
N2	0.12
CO2	0.02
Total	**100**

4.2.10 Hydrogen Specification

The source of hydrogen for the Refinery is provided by the Hydrogen Generation Unit (HGU).

Components	Volume
Hydrogen	99.50 %
Nitrogen	500 ppmv (max.)
Chlorine + Chlorides	1 ppmv (max.)
CO + CO_2	20 ppmv (max.)
CO	10 ppmv (max.)
Water	20 ppmv (max.)
Methane	Balance

4.2.11 Amine Specification

S.No	Parameter	Design
1	Amine Strength	35wt% MDEA
2	Lean amine Loading	0.015 mole of acid gas/ mole of MDEA (max)
3	Rich amine Loading	0.4 mole of acid gas / mole of MDEA (max)

4.2.12 Stripped Sour Water Specification

S.No	Parameter	Design
1	Free H2S	<50 ppmw (max)
2	Free NH3	<50 ppmw (max)

4.2 Economic Aspect

S.No	Utility	Unit	Price (Rs.)
1	Power	Kwh	6.5
2	VHP Steam	MT	3327
3	HP Steam Import	MT	3161
4	MP Steam Import	MT	3003
5	HP Steam Export	MT	2845
6	MP Steam Export	MT	2402
7	LP Steam Import	MT	2823
8	LP Steam Export	MT	0.0
9	Treated Fuel Gas	MT	31942
10	BFW	M3	485
11	Circulating / Cooling water	M3	1.1
12	DM water	M3	86
13	Raw water	M3	3.23
14	Nitrogen	Nm3	12.41
15	Plant Air	Nm3	1.0

4.3 Flare Back Pressure

A new HC flare and acid gas flare system is expected for this expansion project. Maximum flare backpressure shall be considered for sizing of pressure relief devices as follows:

S. No.	Flare system	Superimposed backpressure (kg/cm^2g)	Built-up backpressure at unit battery limits (kg/cm^2g)	Built-up backpressure at PSV outlet (kg/cm^2g)
1	HC Flare (LP Flare)	0.1	1.4	1.7
2	Acid Gas Flare	0.1	0.5	0.7

Notes :

1) Built-up back pressure at PSV outlet shall be estimated based on location of PSV within the unit.
2) Design conditions of new flare networks.

HC Flare:

Design Pressure 3.5 kg/cm2(g)

Design Temperature 320 °C (Hold) (Fire cases are not considered)

Acid Gas Flare:

Design Pressure 3.5 $kg/cm^2(g)$

Design Temperature 200 °C (Hold)